

2

In this chapter, we will learn about

- Kepler's laws of planetary motion
- the apparent weightlessness as experienced by an orbiting body
- the gravitational potential energy of a celestial body
- the escape velocity from the surface of a celestial body

Motions of Celestial Bodies under Gravity

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2.1 Laws of planetary motion

1 Kepler's laws of planetary motion

DSE goal

Describe planetary motion using Kepler's laws.

DSE exam

12(2)Q1.5, 16(2)Q1.4, 20(2)Q1.2, 22(2)Q1.4, 23(2)Q1.6

Flipped classroom

Watch a video and answer the questions.

Johannes Kepler was an astronomer at about the same time as Galileo. He broke the main assumption of ancient astronomy that planets should move in circular orbits. Abandoning the idea of circular orbits, he eventually found great simplicity in the motions of planets. His findings are summarized in **Kepler's laws of planetary motion**.

Know more

Johannes Kepler (1571–1630)

Kepler was an astronomer as well as a mathematician. He was the assistant to another astronomer *Tycho Brahe*, who measured the positions of the planets almost every day. Kepler helped analyze the data and tried to find the laws governing the data. Finally, he published his three laws of planetary motion between 1609 and 1619.

OUP eResources



Simulation

→ This simulation illustrates the meaning of Kepler's first and second laws of planetary motion. It is available on OUP web in PC version (.exe) only.

a Kepler's first law

Kepler's first law tells us the position of the Sun and the shape of the planetary orbit. It states that (Fig 2.1a):

The planets move in elliptical orbits with the Sun at one focus.

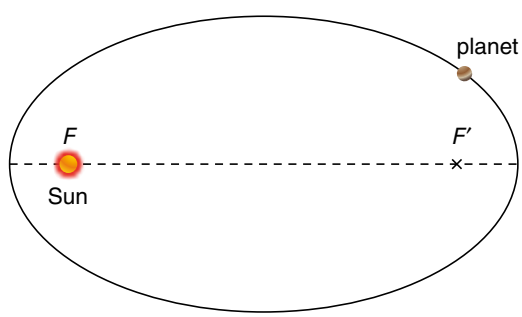


Fig 2.1a *Kepler's first law.*

An elliptical orbit can give a more accurate prediction of the planetary positions. An ellipse is a flattened circle with two foci, F and F'. You will see how an ellipse is drawn in the following activity.

Johannes Kepler 開普勒　　Kepler's laws of planetary motion 開普勒行星運動定律
Tycho Brahe 第谷　　elliptical orbit 橢圓軌道　　focus (pl. foci) 焦點

Video

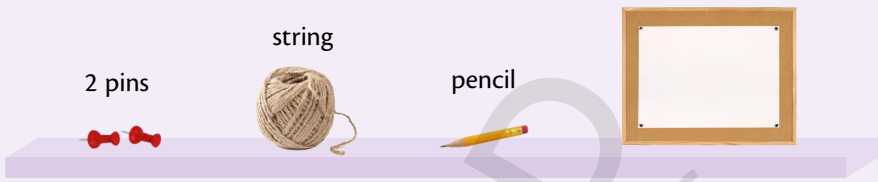
→ This video demonstrates the Hands-on activity.

Hands-on activity

→ Worksheet is available on OUP web.

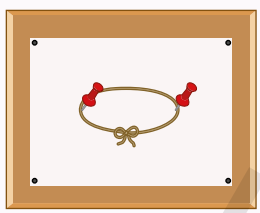
Drawing ellipses

What you need

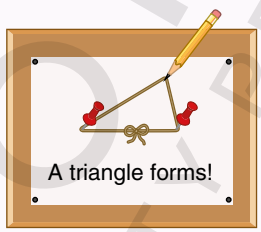


What to do

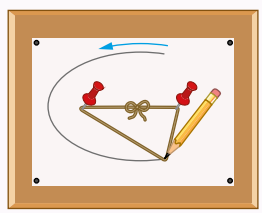
1 Press two pins into the board. Loop a string around the pins.



2 Pull the string taut with a pencil.



3 Move the pencil to draw on the paper while keeping the string taut.



4 Change the distance between the pins. Repeat steps 1–3.
5 Change the length of the string. Repeat steps 1–3.
6 Place the two pins at one point (i.e. the pins coincide). Repeat steps 1–3.

What you find

What shapes can you obtain?

In above activity, the pins represent the positions of the foci. The line joining the two furthest points on an ellipse is called the *major axis*. Half of this line is the *semi-major axis*. The length of the semi-major axis is denoted by a (Fig 2.1b).

In common usage, semi-major axis also represents its length, just like the word 'radius' representing the line and its length.

If the two foci coincide, the ellipse becomes a circle. A circle is a special type of ellipse with its foci, F and F', at the centre and its radius r equals to the semi-major axis a (Fig 2.1c).

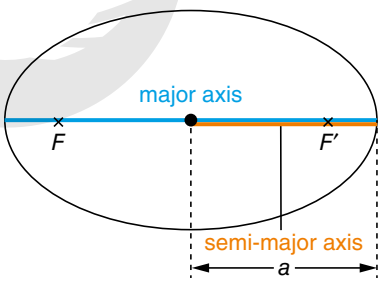


Fig 2.1b *Semi-major axis of an ellipse.*

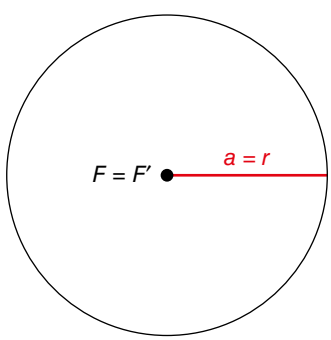


Fig 2.1c *A circle is a special case of an ellipse.*

major axis 長軸　　semi-major axis 半長軸

Earth's orbit is elliptical and the Sun is not at its centre. Therefore, the Earth is sometimes closer and sometimes further away from the Sun.

Kepler's first law points out that a planet is not always at the same distance from the Sun. Sometimes it is closer to the Sun, sometimes further away. The point closest to the Sun in the orbit of a planet is called the *perihelion* and the point furthest from the Sun is called the *aphelion* (Fig 2.1d).

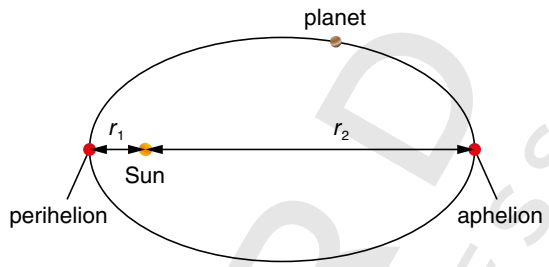


Fig 2.1d *Perihelion and aphelion.*

If the distances of perihelion and aphelion from the Sun are equal to r_1 and r_2 respectively, the semi-major axis a can be found by:

$$a = \frac{1}{2} \times (r_1 + r_2)$$

Know more

More characteristics of an ellipse

An ellipse also has the following characteristics:

1 For all points on an ellipse, the sum of the distances to the foci is a constant (Fig a). (As shown in **Hands-on activity** on p.47, the length of the string is fixed.)

2 The separation between the foci affects how flattened an ellipse is (Fig b).

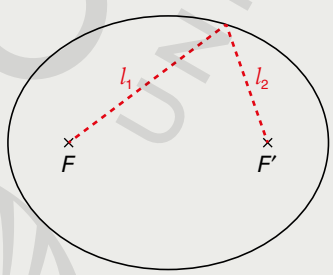


Fig a

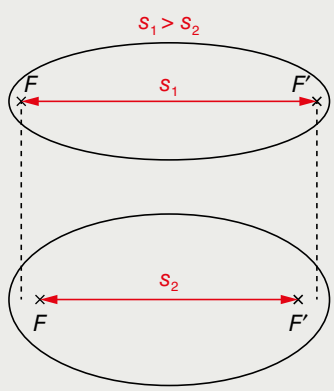


Fig b

b Kepler's second law

Kepler's second law tells us about the speed of a planet as it moves along an elliptical orbit. It states that:

A line from the Sun to any given planet sweeps out equal areas in equal time intervals.

perihelion 近日點　　aphelion 遠日點

For example, in Figure 2.1e, if the planet traces the arcs AB and CD in equal time intervals, the two shaded areas must be equal.

This law implies that **the planet moves faster when it is closer to the Sun**.

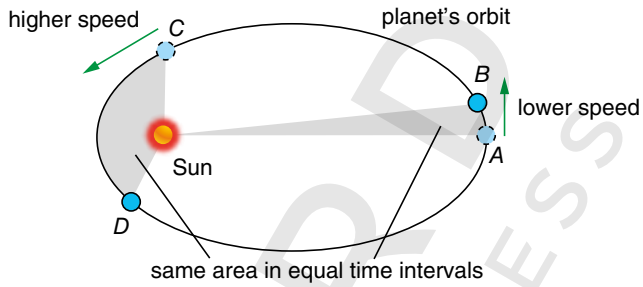


Fig 2.1e *Kepler's second law.*

It also implies that the planet takes half of its orbital period T to move from perihelion to aphelion or from aphelion to perihelion (Fig 2.1f).

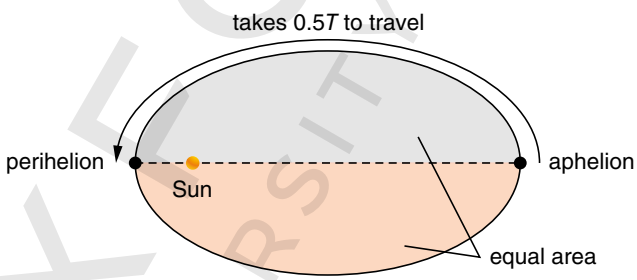


Fig 2.1f *Half of the orbital period is taken to move between the two ends of the major axis.*

Google Forms Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 *True or false:*

(a) The semi-major axis of an elliptical orbit is equal to the distance between its perihelion and aphelion. (T/F)

(b) When a planet gets closer to the Sun, its orbital speed increases. (T/F)

(c) The Sun is at the centre of the Earth's orbit. (T/F)

2 The elliptical orbit of a planet around the Sun is shown below (Fig a).

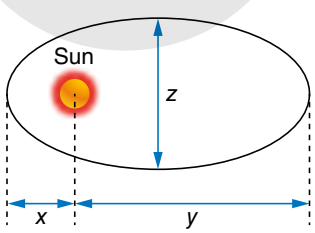


Fig a

What is the semi-major axis of the orbit?

A $\frac{x+y}{2}$ B $\frac{z}{2}$

C $x+y$ D z

3 A planet moves around the Sun in the elliptical orbit shown in Figure b. It passes points A, B and C during its motion. At which point does the planet move the fastest?

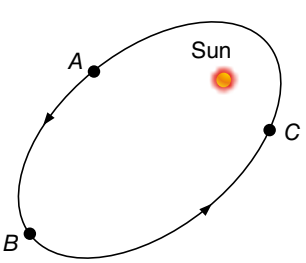


Fig b

C Kepler's third law

Kepler's third law compares the motion of different planets orbiting the Sun in terms of the orbital period and the semi-major axis. It states that:

The square of a planet's orbital period T is proportional to the cube of its semi-major axis a of its orbit, i.e.

$$T^2 \propto a^3 \quad \text{or} \quad \frac{T^2}{a^3} = \text{constant}$$

This law implies that a planet with a longer semi-major axis, which usually means it is further away from the Sun, takes a longer time to orbit the Sun once. For example, Mercury (the closest planet to the Sun) orbits the Sun once in 0.241 year while Neptune (the furthest planet) orbits in 165 years.

Note that Kepler's second law compares the speeds of a planet at different points **in a particular orbit**, while Kepler's third law compares the motions of different planets in **different orbits**.

If we plot a graph of T^2 against a^3, we will get a straight line passing through the origin (Fig 2.1g).

Ts may tell Ss that the graph in Figure 2.1g is plotted on a log-log scale.

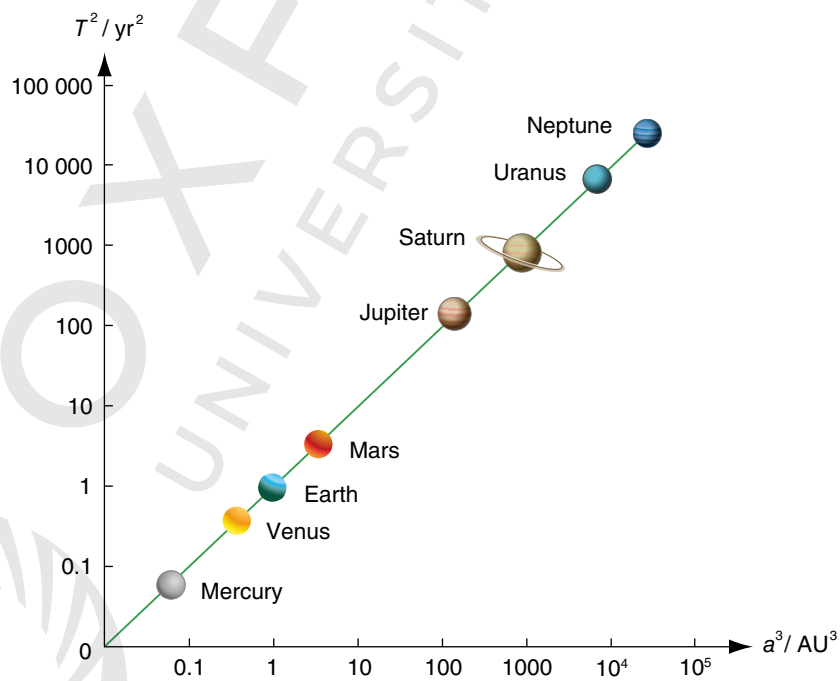


Fig 2.1g *A plot of T^2 against a^3 for planets orbiting the Sun.*

Know more

Kepler's law and the heliocentric model

The work of Kepler built on Copernicus' idea of the heliocentric universe. The discovery that planets move in elliptical orbits suggested that elliptical orbits instead of circular orbits should be used in the Copernican model. The positions of the planets can be predicted accurately if elliptical orbits are used.

Example 1 | Period of Saturn

Saturn moves along an elliptical orbit around the Sun. Its perihelion is 1.35×10^{12} m from the Sun and its aphelion is 1.51×10^{12} m from the Sun.

(a) Find the semi-major axis of Saturn.

(b) The semi-major axis of the Earth is 1 AU (= 1.50×10^{11} m). Find the orbital period of Saturn.

Solution

(a) Semi-major axis $a = \frac{1}{2} \times (1.35 \times 10^{12} + 1.51 \times 10^{12}) = 1.43 \times 10^{12}$ m

(b) Consider Saturn and the Earth. By Kepler's third law,

$$\frac{T^2}{T_{\text{Earth}}^{\ 2}} = \frac{a^3}{a_{\text{Earth}}^{\ 3}}$$

$$\frac{T^2}{1^2} = \frac{(1.43 \times 10^{12})^3}{(1.50 \times 10^{11})^3}$$

$$T = 29.4 \text{ years}$$

Alternative method:

Semi-major axis $a = \frac{1.43 \times 10^{12}}{1.50 \times 10^{11}} = 9.53$ AU

By Kepler's third law,

$$\frac{T^2}{T_{\text{Earth}}^{\ 2}} = \frac{a^3}{a_{\text{Earth}}^{\ 3}}$$

$$\frac{T^2}{1^2} = \frac{9.53^3}{1^3}$$

$$T = 29.4 \text{ years}$$

Tip
Express both T in years and both a in metres.

Tip
Express both T in years and both a in AU.

In addition to planets, Kepler's laws also apply to other smaller bodies in the solar system that orbit the Sun, such as comets and asteroids.

Kepler's laws describe not only the motion of the planets orbiting around the Sun, but also the motion of any objects orbiting a much more massive celestial body under the influence of gravity, such as a planet and its moons (Fig 2.1h). Note that the proportionality constant in the third law is the same only for objects orbiting the same celestial body.

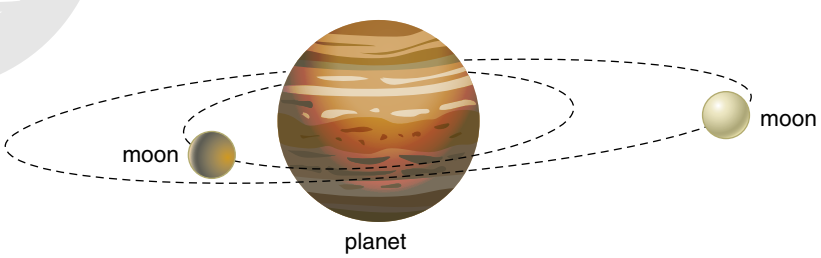
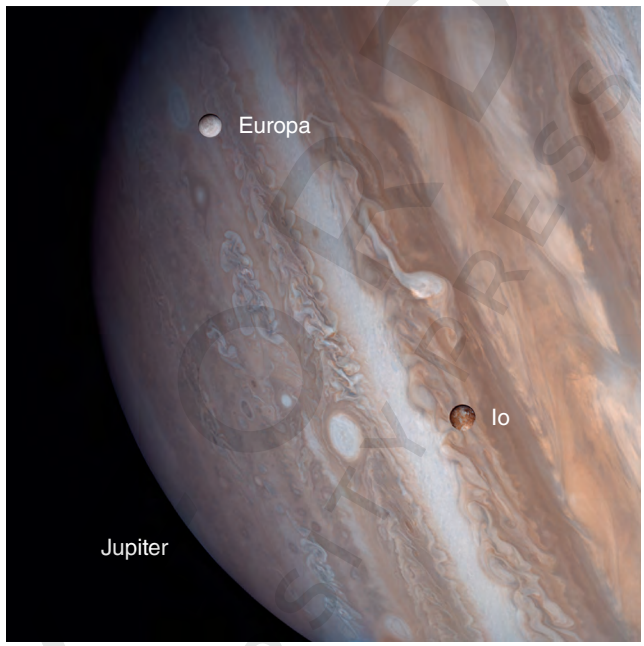


Fig 2.1h *Example in which Kepler's laws are applicable.*

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Example 2 Galilean satellites

The Galilean satellites are the four largest satellites of Jupiter. Io is one of them. The orbital period and semi-major axis of Io are 42.5 hours and 4.22×10^8 m respectively.



(a) The orbit of Io is nearly circular. Estimate the average orbital speed of Io.

(b) Europa is another Galilean satellite. The orbital period of Europa is 3.55 days. Find the semi-major axis of Europa.

(c) The orbital period of the Earth's satellite, the Moon, is 27.3 days. Can we use the above data to estimate the semi-major axis of the Moon by applying Kepler's third law? Explain your answer.

Solution

(a) For a circular orbit, $a = r$.

$$\text{Average orbital speed} = \frac{2\pi r}{T} = \frac{2\pi(4.22 \times 10^8)}{42.5 \times 60 \times 60} = 17\,300 \text{ m s}^{-1}$$

(b) By Kepler's third law,

$$\frac{T_1^2}{T_2^2} = \frac{a_1^3}{a_2^3}$$

$$\frac{(3.55 \times 24)^2}{42.5^2} = \frac{a_1^3}{(4.22 \times 10^8)^3}$$

$$a_1 = 6.71 \times 10^8 \text{ m}$$

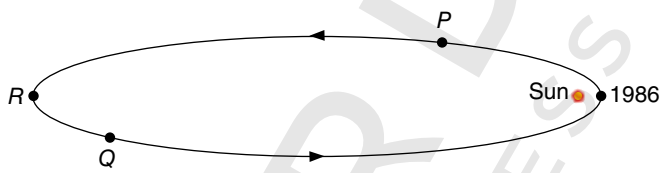
The semi-major axis of Europa is 6.71×10^8 m.

Tip
Express both T in hours and both a in metres.

(c) No, because the Moon is orbiting the Earth, not the same celestial body that Io is orbiting.

Example 3 Comet Halley

Comet Halley orbits the Sun in an elliptical path as shown below. R is the furthest position of Comet Halley from the Sun. P and Q are two positions on its orbit. Take the orbital period of the comet as 75.3 years. The shortest distance between the comet and the Sun is 0.586 AU.



The next appearance of Comet Halley in the night sky will be in the year 2061.

(a) Find the distance x between R and the Sun.

(b) Comet Halley last appeared at the closest position from the Sun in February 1986. Estimate the year in which it returns to R for the first time after that.

(c) Where does Comet Halley have a higher speed, at P or Q? Explain briefly.

Solution

(a) Consider Comet Halley and the Earth. By Kepler's third law,

$$\frac{T^2}{T_{\text{Earth}}{}^2} = \frac{a^3}{a_{\text{Earth}}{}^3}$$

$$\frac{75.3^2}{1^2} = \frac{a^3}{1^3}$$

$$a = 17.83 \text{ AU}$$

By definition of semi-major axis,

$$a = \frac{1}{2} \times (0.586 + x) = 17.83$$

$$x = 35.1 \text{ AU}$$

Tip
Express both T in years and both a in AU.

(b) $\frac{1}{2} \times 75.3 = 37.65$ years

Tip
An orbiting body takes half of its orbital period to move from perihelion to aphelion.

The year Comet Halley first returned to R = 1986 + 37 = 2023

Tip
Comet Halley is at the perihelion in February.
Take this into account,
with $\frac{2}{12} = 0.17$ years,
1986.17 + 37.65 = 2023.82
∴ The answer is 2023.

(c) Comet Halley has a higher speed at P as P is closer to the Sun than Q.

Know more

Comet Halley
Comet Halley has a highly elliptical orbit around the Sun. It is a good example of Kepler's second law.

▸ p.92

Google Forms Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 According to Kepler's third law, a planet in an orbit with a larger semi-major axis would take a (longer / shorter) time to complete one revolution around a star.

2 *True or false:*

(a) Satellites of different masses orbiting a planet with the same semi-major axis have the same orbital period. (T/F)

(b) Kepler's third law only applies to the planets orbiting the Sun. (T/F)

3 The table below lists some celestial bodies orbiting the Sun. They have semi-major axis a and orbital period T. Complete the table. Given that the Earth orbits the Sun once in 365.25 days.

	Celestial body	a / AU	T / days
(a)	A	6.00	
(b)	B		146

4 The table below lists some celestial bodies orbiting the Sun. They have semi-major axis a and orbital period T. Complete the table. Given that the semi-major axis of the Earth's orbit is 1.50×10^{11} m.

	Celestial body	a / m	T / years
(a)	C		2.83
(b)	D	5.00×10^{12}	

5 Two satellites orbit around a planet with circular orbits. If the ratio of their radii is 1 : 2, what is the ratio of their periods?

A 1 : 2

B $1 : 2\sqrt{2}$

C 1 : 4

D 1 : 8

DSE goal

Apply Newton's law of gravitation $F = \frac{GMm}{r^2}$ to explain the motion of celestial bodies in circular orbits.

Derive Kepler's third law $T^2 \propto r^3$ for circular orbits from Newton's law of gravitation.

State Kepler's third law for elliptical orbits $T^2 = \frac{4\pi^2 a^3}{GM}$.

Apply Kepler's third law to solve problems involving circular and elliptical orbits.

DSE exam

17(2)Q.1(a), 21(2)Q1.2, 22(2)Q1.2, 22(2)Q.1(a), 23(2)Q1.3

2 Newton's law of gravitation and Kepler's law

At the beginning of the 16th century, Kepler formulated his three laws of planetary motion by using his and Tycho's data. These laws correctly described many observations and made definite, testable predictions about the future positions of the planets. They provided much greater predictive accuracy than the previous models.

Despite the great success of Kepler's laws, Kepler did not understand why the planets moved or why they followed ellipses. Later in the same century, Newton published his laws of motion which explains Kepler's laws (Fig 2.1i).



Tycho Brahe (1546–1601)
Observation
Data collection

Johannes Kepler (1571–1630)
Analyzation
Model learning

Isaac Newton (1643–1727)
Explanation
Model interpretation

Fig 2.1i *Scientists contributed in studying planetary motion.*

According to Newton, the force acting on a planet due to the Sun has the following properties:

1 its direction is towards the Sun,

2 its magnitude is inversely proportional to the square of the distance from the Sun, and

3 its magnitude is directly proportional to the product of the masses of the planet and the Sun.

The planet governed by this force will eventually move in a way as described by Kepler's laws.

Newton further showed that similar forces are among all celestial bodies (Fig 2.1j). These forces are called gravitational forces and are attractive in nature. The magnitude of the force between two celestial bodies of mass M and m is

$$F = \frac{GMm}{r^2}$$

where $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2} is the universal gravitational constant and r is the distance between the centre of gravity of the two bodies.

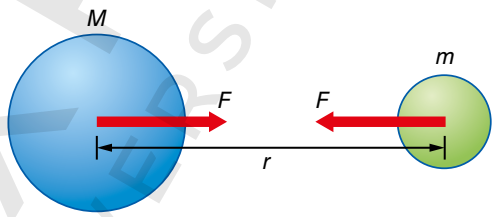


Fig 2.1j *The gravitational force F between two bodies.*

When one body is much more massive than the other, for example $M \gg m$, Newton showed that the lighter body may move around the heavier body in different shapes of paths under the gravitational pull of the heavier body. One such path is the ellipse as stated in Kepler's first law (Fig 2.1k).

Since the orbit is not circular, the instantaneous velocity of the planet is generally not perpendicular to the Sun's gravitational force.

The instantaneous velocity is perpendicular to the gravitational force only at the perihelion and the aphelion.

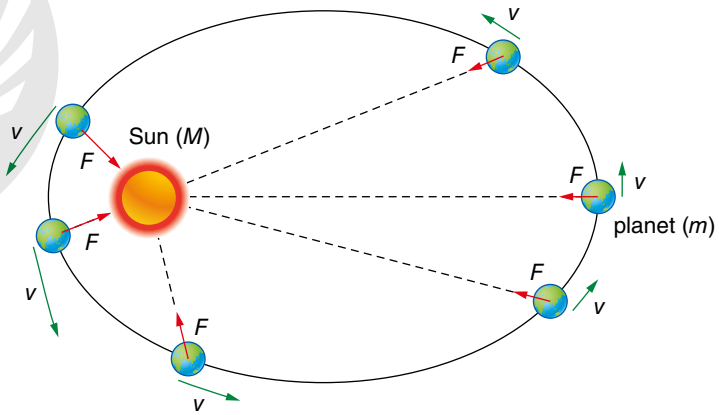


Fig 2.1k *A planet of mass m moving in elliptical orbit around the Sun of mass M is a prediction of Newton's law of gravitation.*

a Derive Kepler's third law for circular motion from Newton's law of gravitation

Assume the orbit is circular. For a planet of mass m orbiting at a constant speed v around the Sun of mass M, the centripetal force required by the planet is provided by the gravitational force between it and the Sun. Therefore,

$$\frac{mv^2}{r} = \frac{GMm}{r^2}$$

$$v^2 = \frac{GM}{r}$$

Since $T = \frac{2\pi r}{v}$,

$$T^2 = \left(\frac{2\pi r}{v}\right)^2 = \frac{4\pi^2 r^2}{v^2} = \frac{4\pi^2 r^2}{\left(\frac{GM}{r}\right)} = \frac{4\pi^2}{GM} r^3 = \text{constant} \times r^3$$

We have

$$T^2 \propto r^3$$

This is Kepler's third law for circular orbits.

Example 4 Artificial satellites around the Earth

An artificial satellite is launched to orbit the Earth at an altitude of 250 km above the Earth's surface.

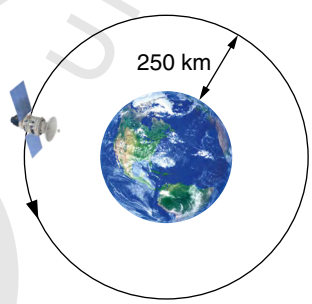


Diagram not drawn to scale

Take the mass and radius of the Earth as 5.97×10^{24} kg and 6370 km respectively. Given: $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}

(a) (i) Find the orbital speed of the satellite.

(ii) If the satellite orbits the Earth at a higher altitude, would the orbital speed be higher or lower than that in (a)? Explain briefly.

(b) (i) Find the orbital period (in hours) of the satellite.

(ii) Another artificial satellite takes twice the time to orbit the Earth once. Estimate its orbital radius.

Solution

(a) (i) As the gravitational force provides the centripetal force,

$$\frac{mv^2}{r} = \frac{GMm}{r^2}$$

$$v = \sqrt{\frac{GM}{r}}$$

$$= \sqrt{\frac{(6.67 \times 10^{-11})(5.97 \times 10^{24})}{(250 + 6370) \times 10^3}} = 7760 \text{ m s}^{-1}$$

The orbital speed of the satellite is 7760 m s^{-1}.

Tip
Remember to convert km to m.

(ii) According to $v = \sqrt{\frac{GM}{r}}$, the orbital speed v would be lower if the orbital radius r is greater.

(b) (i) By Kepler's third law,

$$T^2 = \left(\frac{4\pi^2}{GM}\right)r^3$$

$$T = \sqrt{\left(\frac{4\pi^2}{GM}\right)r^3}$$

$$= \sqrt{\left[\frac{4\pi^2}{(6.67 \times 10^{-11})(5.97 \times 10^{24})}\right][(250 + 6370) \times 10^3]^3}$$

$$= 5360 \text{ s} = 1.49 \text{ h}$$

The orbital period of the satellite is 1.49 hours.

You may also find the orbital period by applying $T = \frac{2\pi r}{v}$. Provided that the calculated value of v must be correct.

(ii) By $T^2 \propto r^3$,

$$\frac{T_1^2}{T_2^2} = \frac{r_1^3}{r_2^3}$$

$$\frac{(2T_2)^2}{T_2^2} = \frac{r_1^3}{[(250 + 6370) \times 10^3]^3}$$

$$r_1 = 1.05 \times 10^7 \text{ m}$$

The orbital radius of the other satellite is 1.05×10^7 m.

Quick check 3

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 Consider satellites moving around the Earth in circular orbits. Determine whether the following statements are true.

(a) The further the satellite is away from the Earth, the greater its orbital speed. (T/F)

(b) For the same orbital radius, a satellite with a larger mass experiences a larger centripetal force, thus moving at a higher orbital speed. (T/F)

(c) Satellites moving with the same period are at the same distance from the Earth. (T/F)

Example 5 Satellites of different planets

Satellite X of mass m orbits around planet A of mass M, with $M \gg m$, in a circular orbit of radius r. Its orbital speed is v and orbital period is T.

(a) Using Newton's law of gravitation, show that $T^2 = \dfrac{4\pi^2 r^3}{GM}$.

(b) Another satellite Y of mass $2m$ orbits around planet B of mass $3M$ in a circular orbit of radius $3r$. Find its period in terms of T.

Solution

(a) As the gravitational force provides the centripetal force,

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$v^2 = \frac{GM}{r}$$

Since $T = \dfrac{2\pi r}{v}$,

$$T^2 = \frac{4\pi^2 r^2}{v^2} = \frac{4\pi^2 r^2}{\left(\dfrac{GM}{r}\right)} = \frac{4\pi^2 r^3}{GM}$$

(b) Since $T^2 = \dfrac{4\pi^2 r^3}{GM}$, $T^2 \propto \dfrac{r^3}{M}$

$$\frac{T_Y^2}{T_X^2} = \frac{r_Y^3}{r_X^3} \times \frac{M_A}{M_B}$$

$$\frac{T_Y^2}{T^2} = \frac{(3r)^3}{r^3} \times \frac{M}{3M}$$

$$T_Y = 3T$$

Period of satellite Y is $3T$.

▶ Exercise 2.1 Q10 (p.62)

b Kepler's third law for elliptical orbits

By using Newton's law of gravitation, Kepler's third law for elliptical orbits ($T^2 \propto a^3$) can be expressed as:

orbital **period** (s) — **semi-major axis** of elliptical orbit (m)

$$T^2 = \left(\frac{4\pi^2}{GM}\right)a^3 = \text{constant} \times a^3$$

universal gravitational constant = 6.67×10^{-11} N m^2 kg^{-2} — **mass** of massive body being orbited (kg)

Note that:

1 All planets that orbit the Sun have the same value for the constant $\frac{4\pi^2}{GM}$, where M is the mass of the Sun. By substituting the values of G and M, we can calculate the constant to be 2.98×10^{-34} yr^2 m^{-3}. This agrees well with the observed data of planets in our solar system (Table 2.1a).

Planets	Orbital period T (yr)	Semi-major axis a ($\times 10^{10}$ m)	T^2/a^3 ($\times 10^{-34}$ yr^2 m^{-3})
Mercury	0.2408	5.791	2.99
Venus	0.6152	10.82	2.99
Earth	1.000	14.96	2.99
Mars	1.881	22.79	2.99
Jupiter	11.86	77.84	2.98
Saturn	29.42	142.7	2.98
Uranus	83.75	287.1	2.96
Neptune	163.7	449.8	2.94

Table 2.1a *Orbital periods and semi-major axes of planets in the solar system.*

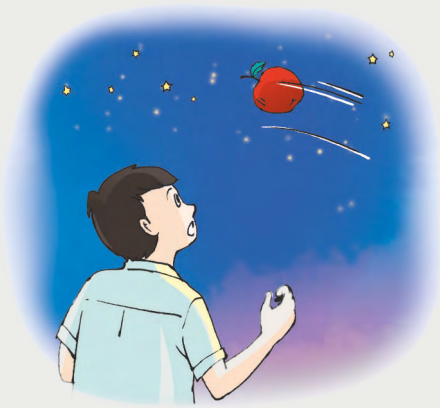
2 Kepler's third law does not only apply to the planets orbiting around the Sun, but also to any celestial body of mass m orbiting around another celestial body of mass M, with $M \gg m$.

In this chapter, unless otherwise specified, assume that the object being orbited is much more massive than the object orbiting around it. ▶

3 The constant $\frac{4\pi^2}{GM}$ depends only on M and does not depend on m.

Know more

The apple, the Moon and Newton's cannonball



Before Newton's time, it was widely believed that the motion of the Moon and the motion of a free-falling apple were caused by different forces. Newton showed that these motions were in fact caused by the same force. How did he explain this?

▶ p.93

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Example 6 Orbital period of Tiangong

The Tiangong space station is in a low Earth elliptical orbit. The closest point and the furthest point are 389.5 km and 395 km above the Earth's surface respectively. Take the mass and radius of the Earth to be 5.97×10^{24} kg and 6370 km respectively. Find the orbital period (in hours) of the Tiangong space station.

Given: $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}

Solution

Semi-major axis $a = \frac{1}{2} \times [(389.5 + 6370) + (395 + 6370)] = 6762$ km

By Kepler's third law for elliptical orbits,

$$T^2 = \left(\frac{4\pi^2}{GM}\right)a^3$$

$$T = \sqrt{\left(\frac{4\pi^2}{GM}\right)a^3}$$

$$= \sqrt{\left[\frac{4\pi^2}{(6.67 \times 10^{-11})(5.97 \times 10^{24})}\right](6762 \times 10^3)^3}$$

$$= 5540 \text{ s} = 1.54 \text{ h}$$

The orbital period of the Tiangong space station is 1.54 hours.

Tip
Divide the value by 3600 to convert it from s to h.

Google Forms Microsoft Forms

Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 A planet orbits a massive star along an elliptical orbit. Which of the following physical quantities would affect the orbital period of the planet? Put a tick in the corresponding box(es).

- ☐ Mass of the planet
- ☐ Mass of the star
- ☐ Semi-major axis of the orbit

2 *True or false:* The gravitational force acting on the Earth by the Sun is always perpendicular to the moving direction of the Earth. (T/F)

3 Some objects orbit different celestial bodies in elliptical orbits. Complete the table below. Given: $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}

	Mass of celestial body	Semi-major axis of orbit	Orbital period
(a)	5.97×10^{24} kg	6680 km	days
(b)	kg	4.22×10^8 m	42.5 hours
(c)	1.99×10^{30} kg	m	75.3 years

Learn by practice 1

Given: $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}

1 The Moon orbits the Earth with a period of 27.3 days. Find the semi-major axis of the Moon's orbit. Take the mass of the Earth to be 5.97×10^{24} kg.

2 The semi-major axis of the Earth's orbit is 1.50×10^{11} m. The Earth has an orbital period of 365.25 days. Find the mass of the Sun.

3 Tianwen-1 orbiter was a spacecraft sent to an orbit around Mars to study the geophysical properties of Mars. Its furthest and closest points are 10 749 km and 275 km above the surface of Mars. The mass and radius of Mars are 6.42×10^{23} kg and 3390 km respectively. Find the orbital period (in hours) of the orbiter.

4 Phobos is a satellite of Mars moving in an elliptical orbit. The semi-major axis of the orbit is 9380 km and the orbital period is 7.66 hours. Find the mass of Mars.

5 A satellite moves in a circular orbit of radius 17 500 km around the Earth. Take the mass of the Earth to be 5.97×10^{24} kg.

(a) Find the orbital speed of the satellite.

(b) Find the orbital period of the satellite.

6 A satellite moves around a planet in a circular orbit. The orbital period is 18 hours. The mass of the planet is 1.24×10^{25} kg.

(a) Find the radius of the orbit.

(b) Find the orbital speed of the satellite.

Exercise 2.1

Take $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}.

★**1** Planets P and Q orbit the same star in circular orbits. The orbital radius of P is 4 times that of Q (Fig a). If T_P and T_Q are the orbital periods of P and Q respectively, what is T_P in terms of T_Q?

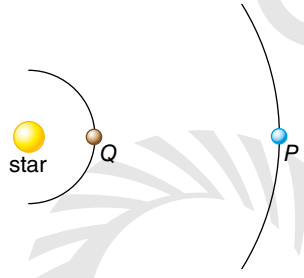


Fig a

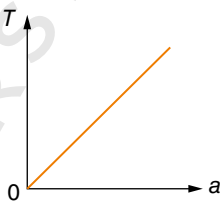
A $4T_Q$

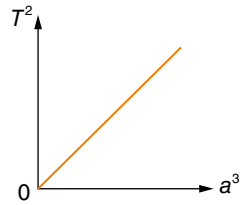
B $8T_Q$

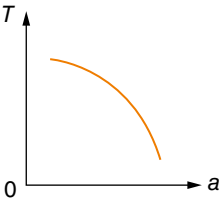
C $16T_Q$

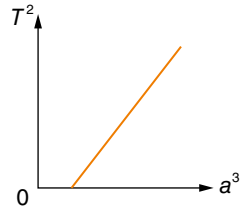
D $64T_Q$

★**2** A planet has 27 natural satellites. Which of the following graphs correctly shows the relationship between the orbital period T and the semi-major axis a of the satellites?

A


B


C


D


★**3** Which of the following can be explained by Kepler's second law?

(1) Comet Halley moves faster when it is closer to the Sun.

(2) A planet moving in a circular orbit has the same speed all the time.

(3) The orbital period of Mercury, the closest planet to the Sun, is shorter than that of the Earth.

A (1) only

B (3) only

C (1) and (2) only

D (1), (2) and (3)

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4 Figure b shows the elliptical path of a celestial body orbiting the Sun. The mass of the Sun is 1.99×10^{30} kg.

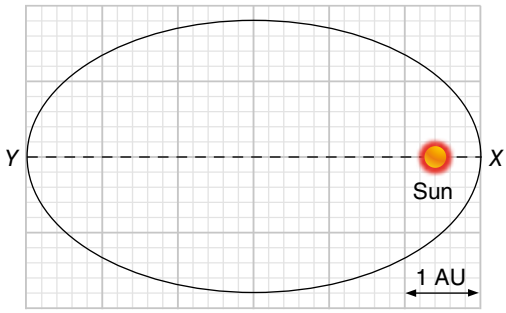


Fig b

(a) X and Y are the points on the orbit which are closest to and furthest away from the Sun respectively. Find the distances of X and Y from the Sun.

(b) Find the semi-major axis of the orbit.

(c) Find the orbital period (in years) of the celestial body.

5 Figure c shows the positions of a planet orbiting around a star at equal time intervals. Mark the approximate position of the star with a cross (×) in the figure.

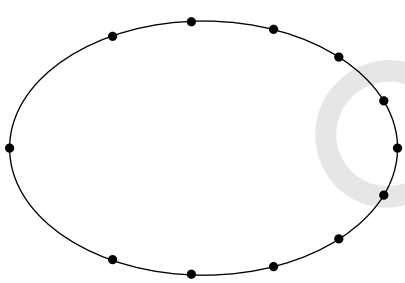


Fig c

★**6** Jupiter's orbital period is 11.8 years. Find the semi-major axis of Jupiter's orbit in metres. Take the semi-major axis of the Earth's orbit to be 1.50×10^{11} m.

★**7** Ganymede is the largest satellite of Jupiter. It completes one revolution around Jupiter in 7.15 days. The semi-major axis of Ganymede's orbit is 1.07×10^{9} m.

(a) Callisto is another satellite of Jupiter with a semi-major axis of 1.88×10^{9} m. Without finding the mass of Jupiter, find the orbital period of Callisto.

(b) Find the mass of Jupiter.

★**8** Makemake is a dwarf planet in the solar system. It completes one revolution around the Sun in 310 years.

(a) The furthest distance of Makemake to the Sun is 52.8 AU. Find its closest distance to the Sun.

(b) (i) State Kepler's second law.

(ii) What does this law suggest about the speed of Makemake during its orbital motion?

★**9** A spacecraft moves in a circular orbit around the Earth. Its orbital speed is 4000 m s^{-1}. Take the mass of the Earth to be 5.97×10^{24} kg.

(a) Find the orbital radius of the spacecraft.

(b) Find the orbital period of the spacecraft.

★**10** Satellite X orbits around the Earth in a circular path of radius r. The orbital period is T. The mass of the Earth is M_E.

(a) Show that $T^2 = \dfrac{4\pi^2 r^3}{GM_E}$ by using Newton's law of gravitation.

(b) Another satellite Y of the same mass orbits the Earth in a circular path at a higher altitude. Explain briefly how each of the following quantities of Y is different from that of X.

(i) Orbital period

(ii) Centripetal force

★**11** The data of four satellites of Saturn are shown in Table a.

Name of satellite	Semi-major axis a ($\times 10^8$ m)	Orbital period T ($\times 10^5$ s)
Mimas	1.85	0.78
Enceladus	2.38	1.21
Tethys	2.95	1.64
Dione	3.77	2.33

Table a

(a) Plot a graph of T^2 against a^3.

(b) Estimate the mass of Saturn from the graph in (a).

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2.2 Weightlessness

1 Apparent weight

Let's consider a girl standing on a weighing scale (Fig 2.2a). The scale reading shows the magnitude of the supporting force acting on her by the balance.

In Book 2 Chapter 3.4, we learned that the girl's weight is the gravitational force acting on her by the Earth. However, it is the supporting force N from the weighing scale that gives her the feeling of her own weight. The greater the supporting force, the more she feels her weight. The supporting force N is her **apparent weight**.

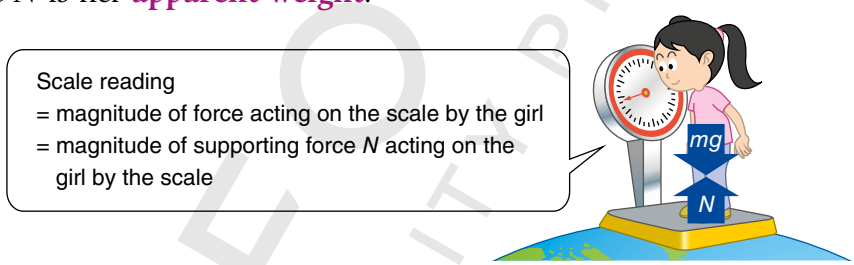


Fig 2.2a *A girl standing on a weighing scale.*

Caution
This result can be obtained by Newton's 1st law. Note that mg and N are not an action-and-reaction pair.

When the girl is stationary, her apparent weight is just equal to her weight ($N = mg$).

When the girl goes down a building in a lift, the lift accelerates downwards as it starts descending (Fig 2.2b(i)). Her apparent weight becomes smaller ($N < mg$) and she feels a loss in weight. However, her weight (mg) does not really change.

$N = mg - ma$

If the lift cable suddenly breaks, the girl and the lift will **fall with the same acceleration g due to gravity** (Fig 2.2b(ii)). The lift floor no longer supports the girl ($N = 0$ as $a = g$). The girl can no longer feel her weight and therefore experiences **weightlessness**, even though her weight does not become zero.

The value of g is independent of the mass of the girl and the lift.

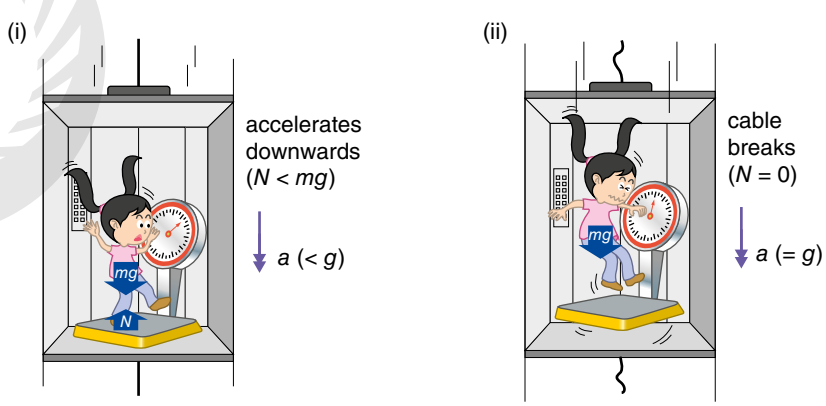


Fig 2.2b *How N changes when (i) the lift accelerates downwards and (ii) the lift cable breaks.*

apparent weight 表觀重量　　weightlessness 失重 / 無重狀態

Figure 2.2c demonstrates this situation. When the weighing scale is released from rest, both the box and the scale fall with the same acceleration g due to gravity. The scale reading becomes zero.



Fig 2.2c *The apparent weight of a box falling freely is zero.*

The weight of the box does not actually become zero and so this kind of weightlessness is called **apparent weightlessness**. This is different from true weightlessness, which occurs only when you are free from gravity, for example, at a position very far away from the Earth and any massive bodies. In that case, your weight truly becomes zero.

The riders of a drop tower may experience a short moment of weightlessness during their fall.



An object experiences apparent weightlessness if the object and the thing that supports it fall with the same acceleration due to gravity.

2 Apparent weightlessness in an orbiting spacecraft

DSE goal

Explain apparent weightlessness in an orbiting spacecraft as a result of acceleration due to gravity being independent of mass.

DSE exam

12(2)Q1.1, 16(2)Q1.1, 17(2)Q1.2

A similar situation occurs in a spacecraft orbiting the Earth. Consider an astronaut (or any object) inside the spacecraft. Take the masses of the Earth, the spacecraft and the astronaut as M, m_S and m_A respectively. The spacecraft is at a distance r from the Earth's centre (Fig 2.2d).

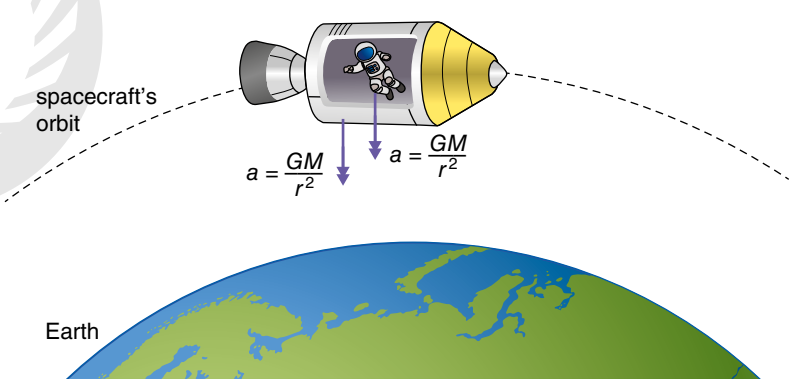
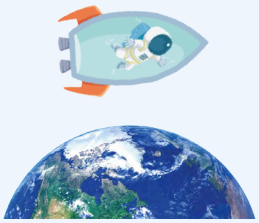


Fig 2.2d *An astronaut in a spacecraft orbiting the Earth.*

Right and wrong



An astronaut experiences weightlessness in a spacecraft orbiting the Earth.

- ✗ The net force acting on the astronaut is zero.
- ✗ The gravitational force acting on the astronaut by the Earth is zero.
- ✓ The normal force acting on the astronaut by the spacecraft is zero.

1 The gravitational forces, i.e. **weights**, acting on the spacecraft and the astronaut are $\frac{GMm_S}{r^2}$ and $\frac{GMm_A}{r^2}$ respectively. They are not the same and are **not zero**.

2 Both the astronaut and the spacecraft 'fall' with the **same acceleration** towards the Earth **due to gravity**. This acceleration is equal to $\frac{GM}{r^2}$, which is **independent of their own masses**. As a result, the astronaut experiences apparent weightlessness. Similarly, other objects inside the spacecraft experience apparent weightlessness and seem to float around (Fig 2.2e).



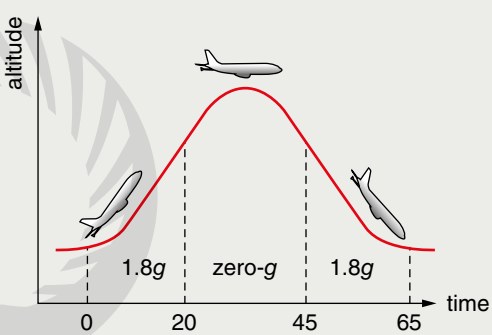


Fig 2.2e *Astronaut and objects float in mid-air in a spacecraft.*

Know more

Reduced-gravity aircraft

A reduced-gravity aircraft can be used to train astronauts and conduct experiments in a weightless environment on Earth. To provide a nearly weightless environment, the aircraft flies in a parabolic path with its thrust reduced. Passengers on the aircraft will experience apparent weightlessness for about 25 s (as shown in the figure).



The aircraft flies up and down in a parabolic path.



The late physicist Stephen Hawking was on the reduced-gravity aircraft in 2007.

Watch a video showing how the passengers experience weightlessness in a reduced-gravity aircraft.

Video

Know more

Writing in space

Let's do a little experiment. Try to write continuously on a notebook with a ballpoint pen upside down while you are lying on bed. You will find that the pen will soon be unable to write because the ink relies on gravity to flow to the tip. How then, can an astronaut write in space?

▸ p.92

Quick check 5

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

1 *True or false:*

(a) A person's weight drops to zero when he experiences apparent weightlessness. (T/F)

(b) Whenever an object near the Earth surface is accelerating with $a = 9.81\ \text{m s}^{-2}$, the object experiences apparent weightlessness. (T/F)

(c) Whenever two objects which are in contact are falling freely under gravity, the objects experience apparent weightlessness. (T/F)

2 Determine whether the following statements about a spacecraft orbiting the Earth are true.

(a) The gravitational force acting on the spacecraft is practically zero. (T/F)

(b) An astronaut inside the spacecraft feels weightlessness only if the orbit of the spacecraft is circular. (T/F)

(c) The accelerations of the astronauts and objects inside the spacecraft are the same as that of the spacecraft. (T/F)

(d) The net forces acting on the astronauts and objects inside the spacecraft are the same as that acting on the spacecraft. (T/F)

Exercise 2.2

1 Patrick holds a scale steady with a box on top of it (Fig a). The scale gives a reading of 20 g. He then drops the scale so that the scale and the box fall freely to the ground.

What is the reading of the scale when the scale and the box are falling freely?

A 0

B 10 g

C 20 g

D 30 g



Fig a

★**2** An astronaut is inside a space station which is orbiting the Earth. Which of the following explains the weightlessness experienced by the astronaut?

A The net force acting on the astronaut is zero.

B The gravitational force acting on the astronaut by the Earth is zero.

C The normal force acting on the astronaut by the space station is zero.

D The normal force and the gravitational force acting on the astronaut balance each other.

★**3** Which of the following astronauts experience(s) weightlessness?

(1) An astronaut in a spacecraft which is just being launched from the Earth's surface

(2) An astronaut in a power-off spacecraft which is travelling in an elliptical orbit around the Earth

(3) An astronaut standing on the Moon's surface

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

★**4** Explain why an astronaut in a spacecraft orbiting a celestial body would experience weightlessness.

★**5** A spacecraft of mass m orbits the Sun in an elliptical orbit. The mass of the Sun is M. At a certain instance, the spacecraft is at a distance r from the Sun.

(a) (i) Write the expression for the magnitude of the spacecraft's acceleration at that moment.

(ii) Is the acceleration of the spacecraft always perpendicular to its instantaneous velocity?

(b) Peter claims that the astronaut in the spacecraft does not feel weightlessness as the orbit is not circular. Comment on this statement.

2.3 Conservation of energy in orbital motions

1 Gravitational potential energy

From now on, gravitational potential energy will be denoted by U.

In Book 2 Chapter 6, we learned that the expression for the change in gravitational potential energy (ΔU) can be found by considering the work done by an external force against the gravitational force.

Consider an object X of mass m near the Earth's surface (Fig 2.3a). If it is lifted by an external force with magnitude mg from a height h_1 to a height h_2 at a constant velocity, its change in PE is given by

In this chapter, PE simply refers to gravitational PE.

$$\Delta U = \text{work done by the external force} = mg(h_2 - h_1) = mg\Delta h$$

DSE goal

Interpret the meaning of gravitational potential energy and its expression $U = -\frac{GMm}{r}$.

By taking the PE at ground level as zero, the PE of X at any height h above the ground level can be written as

$$U = mgh$$

DSE exam

15(2)Q1.1, 18(2)Q1.1

However, this only applies to the case **near** the Earth's surface where g is approximately constant. If the change in height is large, g is no longer constant and we have to use a more general expression for the PE.

Assume that object Y is fixed in position.

Again, consider an object X of mass m initially at position P. It is attracted by another object Y of mass M due to gravity, but it is moved by an external force from position P to position Q at a constant velocity (Fig 2.3b).

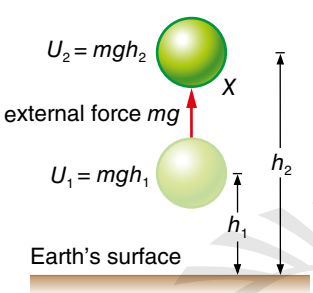


Fig 2.3a *Moving an object from h_1 to h_2.*

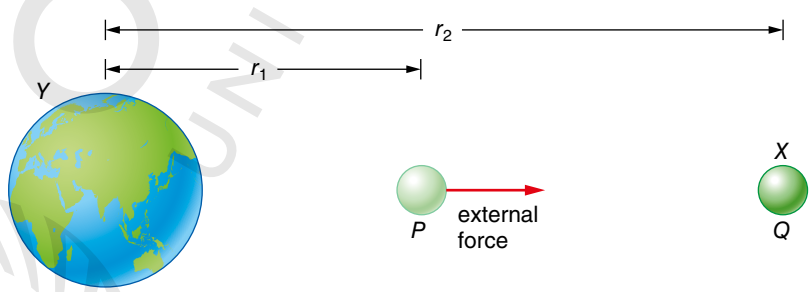


Fig 2.3b *Moving object X from position P to position Q.*

The change in PE (ΔU) of X is equal to the work done by the external force. Since the external force is in the same direction as the displacement, the work done by this force is positive (i.e. $\Delta U > 0$).

Unless otherwise specified, PE at infinity is taken as zero in this chapter.

By taking the PE at infinity as zero, and considering the case when X is moved from a finite distance r to infinity,

$$\Delta U = U_\infty - U_r = 0 - U_r$$

Therefore, the PE at any finite distance r from Y must be negative ($U_r = -\Delta U < 0$).

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The derivation is out of the scope of this book.

We can derive the expression of gravitation PE as follows:

Consider the work done by gravitation force.

$dU = \frac{GMm}{r^2}\,dr$

Integrate both sides and take $U = 0$ at infinity:

$U = -\frac{GMm}{r}$

From Newton's law of gravitation and taking the PE at infinity as zero, the PE of X at any distance r from Y can be expressed as

universal gravitational constant $= 6.67 \times 10^{-11}$ N m^2 kg^{-2}

mass of Y (kg)

PE of a X (J)

$$U = -\frac{GMm}{r}$$

mass of X (kg)

separation between the c.g. of the two bodies (m)

The PE of X is the energy that it possesses in relation to Y due to gravity. This energy can be converted into KE if X is released from rest.

Note that the further away from Y, the larger the PE that X possesses (i.e. when r increases, U becomes less negative). Figure 2.3c shows how U varies with r.

R is the radius of object Y.

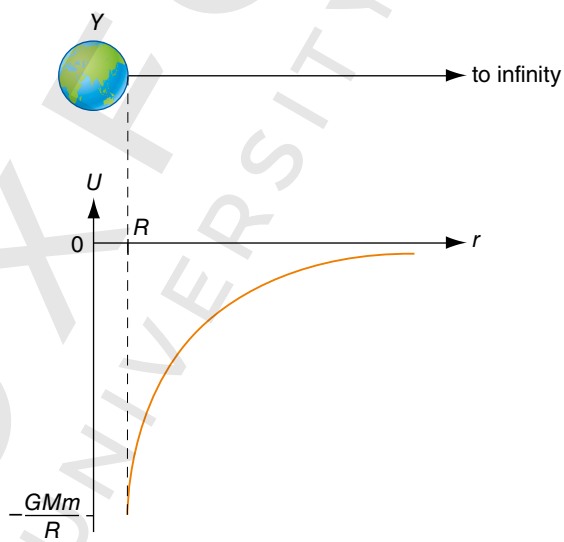


Fig 2.3c *Variation of U against r.*

In Figure 2.3b on p.67, X is moved from position P (r_1 from Y) to position Q (r_2 from Y). Its change in PE is

$$\begin{aligned}\Delta U &= U_Q - U_P \\ &= -\frac{GMm}{r_2} - \left(-\frac{GMm}{r_1}\right) \\ &= GMm\left(\frac{1}{r_1} - \frac{1}{r_2}\right)\end{aligned}$$

This equation can be applied no matter $r_1 > r_2$ or $r_1 < r_2$. If $r_1 > r_2$, X is moved to a position closer to Y. X loses PE and ΔU is negative. If $r_1 < r_2$, X is moved to a position further away from Y. X gains PE and ΔU is positive.

Example 7 | Potential energy due to the Earth

A satellite of mass 50 000 kg is launched from the Earth's surface to 850 km above the ground. Take the mass and radius of the Earth to be 5.97×10^{24} kg and 6370 km respectively. Given: $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}

(a) Find the potential energy U of the satellite when it is at 850 km above the ground.

(b) Find the change in potential energy ΔU of the satellite during the launch.

(c) Consider a general case when a satellite of mass m is launched from the Earth's surface to a height of h above the Earth's surface. Given that the radius of the Earth is R and the acceleration due to gravity on the Earth's surface is g, show that the change in potential energy ΔU of the satellite is $mgh\left(\frac{R}{R+h}\right)$.

Solution

(a) $U = -\frac{GMm}{r}$

$= -\frac{(6.67 \times 10^{-11})(5.97 \times 10^{24})(50\ 000)}{(850 + 6370) \times 10^3}$

$= -2.76 \times 10^{12}$ J

r is the distance between the c.g. of the objects. Therefore, the radius of the Earth should be included.

(b) ΔU

$= -\frac{GMm}{r_2} - \left(-\frac{GMm}{r_1}\right)$

$= GMm\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$

$= (6.67 \times 10^{-11})(5.97 \times 10^{24})(50\ 000)\left[\frac{1}{6370 \times 10^3} - \frac{1}{(850 + 6370) \times 10^3}\right]$

$= 3.68 \times 10^{11}$ J

(c) $\Delta U = -\frac{GMm}{R+h} - \left(-\frac{GMm}{R}\right)$

$= GMm\left(\frac{1}{R} - \frac{1}{R+h}\right)$

$= GMm\left[\frac{h}{R(R+h)}\right]$

$= \left(\frac{GM}{R^2}\right)\left(\frac{mRh}{R+h}\right)$

$= g \times \left(\frac{mRh}{R+h}\right)$

$= mgh\left(\frac{R}{R+h}\right)$

This shows that the equation $\Delta PE = mg\Delta h$ is only valid when an object has a small change in height near the Earth's surface.

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Google Forms Microsoft Forms

Quick check 6

→ E-version in OUP Exercise Platform is also available.

1 A spacecraft of mass m is at a distance d above the ground of a planet of mass M. Which of the following situations would increase the gravitational potential energy of the spacecraft? Put a tick in the corresponding box(es).

- ☐ d increases.
- ☐ m increases.
- ☐ M increases.

2 *True or false:* According to $U = -\frac{GMm}{r}$, an object has a larger U when it is on the ground than on a rooftop. (T/F)

3 Find the gravitational potential energy U of a 50-kg person at the following locations by applying $U = -\frac{GMm}{r}$. The mass and radius of the Earth are 5.97×10^{24} kg and 6370 km respectively. Given: $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}

	Location	U / J
(a)	At the Earth's surface	
(b)	At a height of 500 km above the Earth's surface	
(c)	At infinity	

Learn by practice 2

Take $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}.

1 An astronaut of mass 60 kg stands on the Moon's surface. Find his gravitational potential energy. The mass and radius of the Moon are 7.35×10^{22} kg and 1.74×10^{6} m respectively.

2 The mass and radius of the Earth are 5.97×10^{24} kg and 6370 km respectively. Find the change in gravitational potential energy of a spacecraft with a mass of 12 000 kg in each process below.

(a) Launching from the Earth's surface to a height of 1800 km

(b) Transferring from an orbit of radius 8170 km to an orbit of radius 7500 km

2 Conservation of energy

DSE goal

Apply conservation of mechanical energy to solve problems involving the motion of celestial bodies or spacecraft.

DSE exam

13(2)Q.1, 19(2)Q.1(a), 22(2)Q1.3, 22(2)Q.1(b), 23(2)Q1.5

If you drop a ball from your hand, it loses PE and gains KE. The total mechanical energy of the ball is conserved. This is also true for orbital motions for celestial bodies.

$$K_1 + U_1 = K_2 + U_2$$

KE can be denoted by K.

Note that U is no longer given by mgh but $-\frac{GMm}{r}$.

a For circular orbits

Consider a satellite moving around the Earth in a circular orbit with orbital radius r and orbital speed v. The gravitational force between the satellite and the Earth provides the centripetal force for the circular motion of the satellite. We have

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$v = \sqrt{\frac{GM}{r}}$$

Therefore,

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\sqrt{\frac{GM}{r}}\right)^2 = \frac{GMm}{2r}$$

Then, the total mechanical energy of the satellite is

$$K + U = \frac{GMm}{2r} + \left(-\frac{GMm}{r}\right) = -\frac{GMm}{2r} = \frac{1}{2}U \quad \text{(for circular orbits only)}$$

From the equation, we can see that the total mechanical energy of a body orbiting another body in a circular orbit under gravity is always negative.

Example 8 Total mechanical energy in a circular orbit

Hubble Space Telescope (Fig a) has a mass of 12 200 kg. Assume that it orbits the Earth at a fixed height of 539 km above the Earth's surface. Find its total mechanical energy in its orbit. Take the mass and radius of the Earth to be 5.97×10^{24} kg and 6370 km respectively.
Given: $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}



Fig a

Solution

By $\frac{GMm}{r^2} = \frac{mv^2}{r}$,

$$v = \sqrt{\frac{GM}{r}}$$

Total mechanical energy

$$= \frac{1}{2}mv^2 + \left(-\frac{GMm}{r}\right)$$

$$= \frac{1}{2}m\left(\sqrt{\frac{GM}{r}}\right)^2 + \left(-\frac{GMm}{r}\right)$$

$$= \frac{GMm}{2r} + \left(-\frac{GMm}{r}\right)$$

$$= -\frac{GMm}{2r}$$

$$= -\frac{(6.67 \times 10^{-11})(5.97 \times 10^{24})(12\,200)}{2[(539 + 6370) \times 10^3]}$$

$$= -3.52 \times 10^{11}\ \text{J}$$

b For elliptical orbits

Consider a planet moving around a star in an elliptical orbit (Fig 2.3d). It is closest to the star at position A and furthest away at position B.

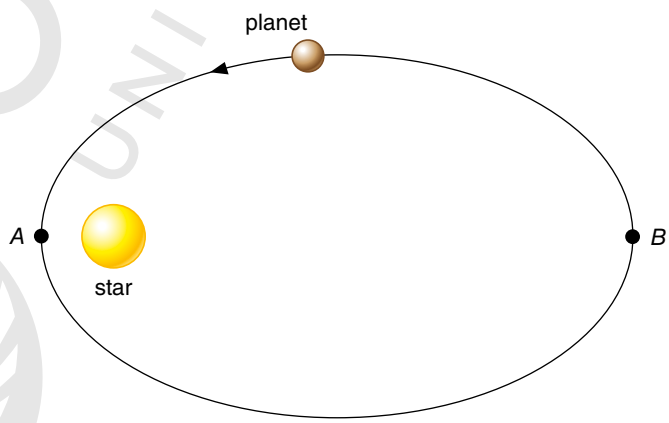


Fig 2.3d *A planet orbiting a star in an elliptical path.*

As the total mechanical energy of the planet is conserved,

$$K_A + U_A = K_B + U_B$$

As U is less negative at position B which is further away from the star, therefore $U_A < U_B$. ▶ Since $U_A < U_B$, we have $K_A > K_B$.

That is, the planet moves faster at position A than at position B. This is in line with Kepler's second law, though we cannot prove it completely at this level.

✎ Kepler's second law is a consequence of the conservation of angular momentum.

Example 9 Orbital speed of a spacecraft

A spacecraft moves around the Earth in an elliptical orbit. The speed of the spacecraft is 1.00×10^4 m s^{-1} when it is at position P which is at 200 km above the Earth's surface. Take the mass and radius of the Earth to be 5.97×10^{24} kg and 6370 km respectively.

Given: $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}

(a) Find the speed of the spacecraft at position Q which is at 50 km above the Earth's surface.

(b) The mass of the spacecraft is 1.2×10^5 kg. Find the total mechanical energy of the spacecraft at position P.

(c) If the spacecraft is powered to move from position P to infinity, what is the minimum energy needed by the spacecraft?

Solution

(a) $K_1 + U_1 = K_2 + U_2$

$$\frac{1}{2}mv_1^2 + \left(-\frac{GMm}{r_1}\right) = \frac{1}{2}mv_2^2 + \left(-\frac{GMm}{r_2}\right)$$

$$\frac{1}{2}v_1^2 + \left(-\frac{GM}{r_1}\right) = \frac{1}{2}v_2^2 + \left(-\frac{GM}{r_2}\right)$$

$$GM\left(\frac{1}{r_2} - \frac{1}{r_1}\right) = \frac{1}{2}(v_2^2 - v_1^2)$$

$$(6.67 \times 10^{-11})(5.97 \times 10^{24})\left[\frac{1}{(50 + 6370) \times 10^3} - \frac{1}{(200 + 6370) \times 10^3}\right] = \frac{1}{2}[v_2^2 - (1.00 \times 10^4)^2]$$

$$v_2 = 1.01 \times 10^4 \text{ m s}^{-1}$$

The speed of the spacecraft at position Q is 1.01×10^4 m s^{-1}.

(b) Total mechanical energy at position P

$$= \frac{1}{2}mv_1^2 + \left(-\frac{GMm}{r_1}\right)$$

$$= \frac{1}{2} \times (1.2 \times 10^5) \times (1.00 \times 10^4)^2 + \left[-\frac{(6.67 \times 10^{-11})(5.97 \times 10^{24})(1.2 \times 10^5)}{(200 + 6370) \times 10^3}\right]$$

$$= -1.27 \times 10^{12} \text{ J}$$

(c) The minimum total energy of the spacecraft at infinity is 0.

$\therefore$ The minimum energy needed is 1.27×10^{12} J.

Caution

The total mechanical energy of the spacecraft is **not** equal to $\frac{1}{2}U$ as its orbit is not circular.

Think about...

If the spacecraft is powered to move from position Q to infinity, what is the minimum energy needed by the spacecraft?

Learn by practice 3

Take $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}.

1 Comet Halley orbits the Sun in an elliptical path. Its shortest and longest distances from the Sun are 8.78×10^{10} m and 5.27×10^{12} m respectively. When it is closest to the Sun, its speed is about 54 550 m s^{-1}. Estimate its speed when it is furthest from the Sun. Take the mass of the Sun to be 1.99×10^{30} kg.

2 An object is projected from the Earth's surface with a launching speed of 4800 m s^{-1}. What is the maximum distance from the Earth's surface that the object can reach? Take the mass and radius of the Earth to be 5.97×10^{24} kg and 6370 km respectively. Neglect air resistance.

3 Find the total mechanical energy of

(a) Comet Halley in Q1, take its mass as 2.2×10^{14} kg;

(b) the projected object in Q2, take its mass as 10 kg.

4 Find the total mechanical energy of an object moving around the Earth in a circular orbit of radius 7500 km. The masses of the object and the Earth are 800 kg and 5.97×10^{24} kg respectively.

DSE goal

Determine the escape velocity on a celestial body.

DSE exam

14(2)Q1.6, 16(2)Q1.2, 17(2)Q1.3, 18(2)Q1.2, 20(2)Q1.3, 21(2)Q1.1, 22(2)Q1.1, 23(2)Q1.4

Simulation

→ This simulation lets Ss launch a rocket at different speeds and find out the escape velocity of different celestial bodies. It is available on OUP web in PC version (.exe) only.

3 Escape velocity

When you throw a ball vertically upward, the ball will go up and come down (Fig 2.3e). If you give the ball a higher initial speed, it will reach a greater height. What is the minimum initial speed required for the ball to escape from the Earth without returning?



Fig 2.3e *Throwing up a ball.*

When an object of mass m is projected with a speed v from the surface of a celestial body of mass M (with $M \gg m$) and radius R,

$$\text{total mechanical energy of the object} = K + U = \frac{1}{2}mv^2 + \left(-\frac{GMm}{R}\right)$$

If the object is given a minimum initial speed to escape from the gravitational pull of the celestial body, it will just reach infinity and its KE will be zero there. Both its KE and PE at infinity are zero, and so is its total mechanical energy (Fig 2.3f).

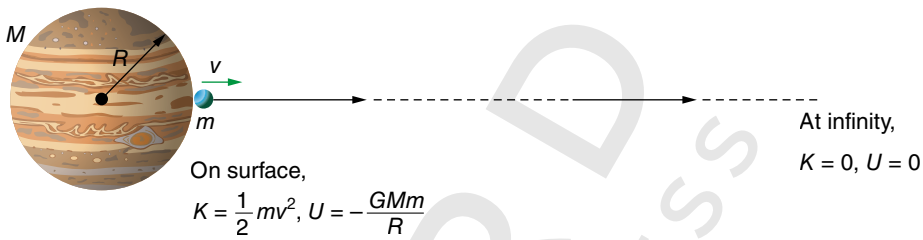


Fig 2.3f *KE and PE of an object just able to escape from a celestial body.*

Note that the air resistance is ignored in the calculation of escape velocity here.

▶ As the total mechanical energy of the object is conserved, we have

$$\frac{1}{2}mv^2 + \left(-\frac{GMm}{R}\right) = 0$$

$$\frac{1}{2}mv^2 = \frac{GMm}{R}$$

$$v = \sqrt{\frac{2GM}{R}}$$

This speed is called the **escape velocity**.

Escape velocity is the minimum speed required for an object on the surface of a celestial body to escape from the gravitational influence of the body.

universal gravitational constant = 6.67×10^{-11} N m^2 kg^{-2} ····· **mass** of celestial body (kg)

escape **velocity** (m s^{-1}) ····· $v_e = \sqrt{\frac{2GM}{R}}$ ····· **radius** of celestial body (m)

Note the following about the escape velocity:

1 It is independent of the mass of the projected objects.

2 The formula applies to objects that are not powered after projection. A powered vehicle (such as a rocket) does not need to achieve this speed to escape from the celestial body to infinity.

3 It is a scalar quantity. An object can be projected in different directions with the escape velocity to escape from the celestial body.

4 The formula applies to objects projected at the surface of the celestial body. If an object is at a distance r from the centre of the celestial body, it could escape with a minimum speed of $\sqrt{\frac{2GM}{r}}$ (with $r > R$).

Know more

The atmosphere

Not all celestial bodies have an atmosphere like the Earth. A celestial body with a low escape velocity may not have an atmosphere. Why?

▶ p.94

escape velocity 逃逸速度

When an unpowered object is projected horizontally near the surface of a celestial body, its subsequent motion is determined by its projecting speed. Figure 2.3g states the critical conditions for different kinds of subsequent motion of such an object.

If the projecting speed increases gradually from the speed for a circular orbit $\left(\sqrt{\frac{GM}{R}}\right)$, the object will move in elliptical orbits with a longer and longer semi-major axis. If the projecting speed surpasses the escape velocity $\left(\sqrt{\frac{2GM}{R}}\right)$, the object will escape from the celestial body.

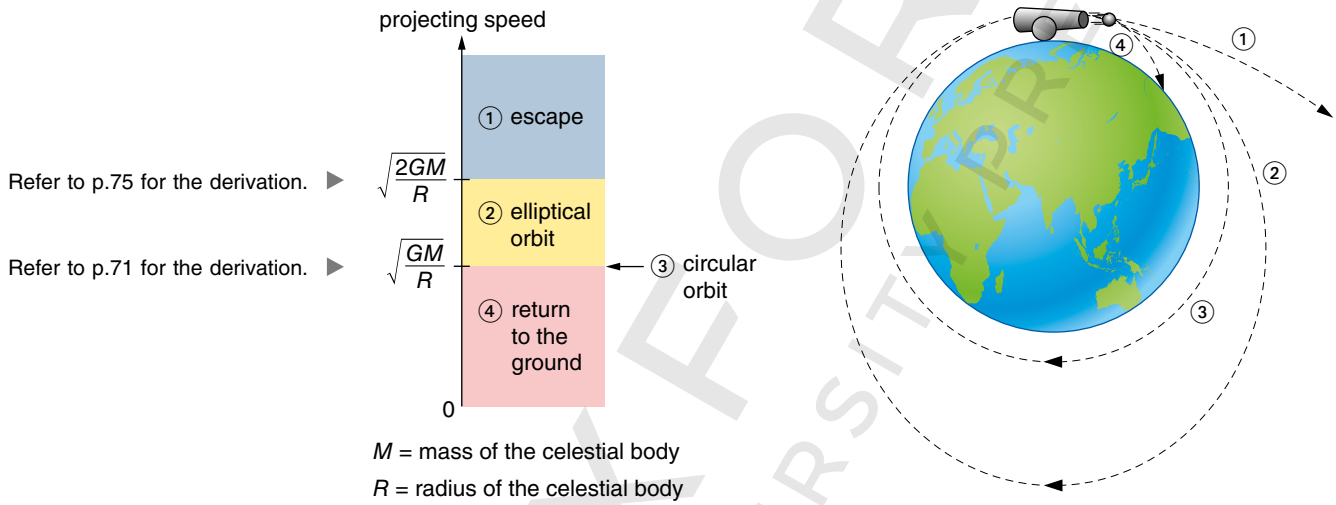


Fig 2.3g *Critical conditions for different kinds of subsequent motion of an object projected horizontally near the surface of a celestial body.*

Example 10 Escape velocity from the Earth

The mass and radius of the Earth are 5.97×10^{24} kg and 6370 km respectively. Find the escape velocity

(a) from the Earth's surface,

(b) from a height of 200 km above the Earth's surface.

Given: $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}

Solution

(a) Escape velocity from the Earth's surface

$$= \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2 \times (6.67 \times 10^{-11}) \times (5.97 \times 10^{24})}{6370 \times 10^3}} = 11\ 200 \text{ m s}^{-1}$$

(b) Escape velocity from a height of 200 km above the Earth's surface

$$= \sqrt{\frac{2GM}{r}} = \sqrt{\frac{2 \times (6.67 \times 10^{-11}) \times (5.97 \times 10^{24})}{(200 + 6370) \times 10^3}} = 11\ 000 \text{ m s}^{-1}$$

Google Forms Microsoft Forms

Quick check 7

→ E-version in OUP Exercise Platform is also available.

1 *True or false:* In order to escape from the Earth's surface, a rocket must keep on firing its engine to maintain its speed fixed at escape velocity. (T/F)

(For Q2–3.) Put a tick in the corresponding box(es).

2 An object leaves the Earth's surface at the escape velocity. When it is infinitely far away from the Earth,

- ☐ its PE is zero.
- ☐ its KE is zero.
- ☐ its speed is zero.

3 An object leaves the Earth's surface at a speed higher than the escape velocity. When it is infinitely far away from the Earth,

- ☐ its PE is zero.
- ☐ its KE is zero.
- ☐ its speed is zero.

4 Arcturus is a red giant star located in the constellation of Bootes. Its mass and radius are 2.15×10^{30} kg and 1.77×10^{10} m respectively. Find the escape velocity at its surface. Take $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}.

5 A object is projected at speed v horizontally from a position near the Earth's surface (Fig a). Take M and R as the mass and radius of the Earth respectively. Determine the subsequent motion of the object in each of the following cases by choosing between A (eventually return to the Earth), B (fly in a circular orbit), C (fly in a non-circular orbit) and D (escape from the Earth).

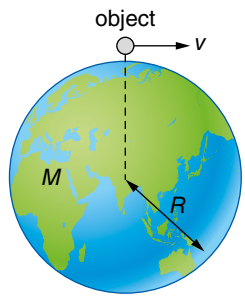


Fig a

(a) $v = \sqrt{\frac{GM}{R}}$

(b) $\sqrt{\frac{GM}{R}} < v < \sqrt{\frac{2GM}{R}}$

(c) $v = \sqrt{\frac{2GM}{R}}$

(d) $v > \sqrt{\frac{2GM}{R}}$

6 What will the answers in Q5 be if the object is projected vertically upwards (Fig b)?

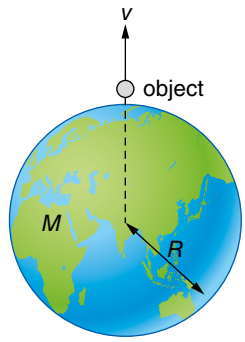


Fig b

Writing practice

Question

A spacecraft initially orbits around the Earth in a circular orbit.



At a certain position, it fires its engine momentarily to boost up its speed. The spacecraft still orbits the Earth after this operation. Explain how the orbital period of the spacecraft changes. (2 marks)

Answer

Refer to p.76 for detailed explanation. ▶

After the increase in speed, the spacecraft moves in an elliptical orbit with a longer semi-major axis (longer than the radius of the circular orbit). 1A

As $T^2 \propto a^3$, the orbital period of the spacecraft becomes longer. 1A

Practise yourself

1 A spacecraft orbits around a celestial body in a circular orbit. Raymond suggests that the spacecraft can fire its rocket momentarily and then move in the same orbit at a higher speed. Comment on his suggestion. (2 marks)

☆ HKDSE 2018 Paper 2 Q1.2

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Speed at infinity

The escape velocity from the Earth's surface is x. If a space probe is released from the Earth's surface at a speed of $3x$, what is the speed of the space probe when it is very far away from the Earth?

A 0

B x

C $\sqrt{8}x$

D $\sqrt{10}x$

Solution

If the space probe is released at escape velocity x,

$$K + U = K_\infty + U_\infty$$

The space probe can be considered at infinity when it is very far away from the Earth.

$$\frac{1}{2}mx^2 + \left(-\frac{GMm}{R}\right) = 0 + 0$$

$$\frac{1}{2}mx^2 = \frac{GMm}{R}$$

If the space probe is released at speed $3x$,

$$K + U = K_\infty + U_\infty$$

$$\frac{1}{2}m(3x)^2 + \left(-\frac{GMm}{R}\right) = \frac{1}{2}mv^2 + 0$$

Common mistake ⚠
KE at infinity is not zero when an object is not projected with the escape velocity.

$$\frac{1}{2}m(3x)^2 - \frac{1}{2}mx^2 = \frac{1}{2}mv^2$$

$$9x^2 - x^2 = v^2$$

$$v = \sqrt{8}x$$

∴ The answer is C.

▶ Revision 2 Q16 (p.86)

☆ HKDSE 2013 Paper 2 Q.1, HKDSE 2019 Paper 2 Q.1

→ This video demonstrates how this question is solved.

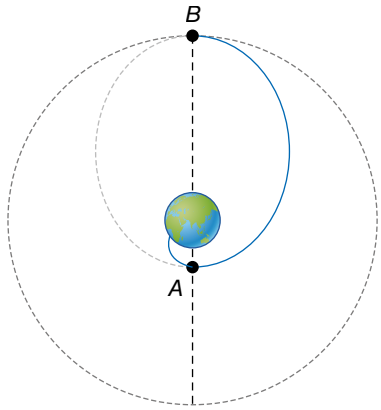
DSE practice 2 | Launch of a geostationary satellite

Video

GOES-18 is a satellite for continuously monitoring the weather in Western Hemisphere. When it was launched in March 2022, it was first brought by a rocket to position A and was then separated from the remaining part of the rocket. It then moved in an elliptical transfer orbit from A to B with its engine shut down and entered a geostationary orbit at B. AB is the major axis of the transfer orbit.

Take the mass and radius of the Earth to be 5.97×10^{24} kg and 6370 km respectively.

Given: $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}



(a) Suggest and explain one advantage for the satellite to move in the geostationary orbit. (2 marks)

(b) Find the radius of the geostationary orbit. (2 marks)

(c) Position A is at about 2530 km above the Earth's surface. Estimate the time (in min) for the satellite to travel from A to B. (3 marks)

(d) Suggest a way for the satellite to switch from the elliptical transfer orbit to the geostationary orbit at B. (1 mark)

(e) The total mechanical energy and the gravitational potential energy of the satellite in the geostationary orbit are E and U respectively. Show that $E = \frac{1}{2}U$. (2 marks)

Solution

(a) The satellite remains directly above a certain location on the Earth. 1A

It can therefore monitor the weather of the same location continuously. 1A

(b) By Kepler's third law,

$$T^2 = \left(\frac{4\pi^2}{GM}\right) r^3 \quad \text{1M}$$

$$(24 \times 3600)^2 = \left[\frac{4\pi^2}{(6.67 \times 10^{-11})(5.97 \times 10^{24})}\right] r^3$$

$$r = 4.22 \times 10^7 \text{ m} \quad \text{1A}$$

The radius of the geostationary orbit is 4.22×10^7 m.

Remember to use SI units for this formula.

The orbital period of the geostationary orbit (i.e. the rotation period of the Earth) is actually 23 h 56 min 4 s.

(c) For the transfer orbit,

$$a = \frac{1}{2} \times [(2530 + 6370) \times 10^3 + 4.22 \times 10^7)] = 2.555 \times 10^7 \text{ m} \quad \text{1M}$$

Common mistake
Students may forget to include the radius of the Earth in calculating the distance of A from the Earth's centre. They may also forget how to calculate the semi-major axis of an elliptical orbit.

By Kepler's third law,

$$T^2 = \left(\frac{4\pi^2}{GM}\right) a^3$$

$$T = \sqrt{\left[\frac{4\pi^2}{(6.67 \times 10^{-11})(5.97 \times 10^{24})}\right](2.555 \times 10^7)^3} \quad \text{1M}$$

$$= 40\,700 \text{ s}$$

Time for the satellite to travel from A to B

$$= \frac{1}{2} \times 40\,700 = 20\,350 \text{ s} = 339 \text{ min} \quad \text{1A}$$

Common mistake
Students may forget that the satellite only takes half of the period to move from A to B in the transfer orbit.

(d) Fire its engine momentarily to boost up its speed at B to the required orbital speed for the geostationary orbit. 1A

(e) By $\frac{GMm}{r^2} = \frac{mv^2}{r}$, $v = \sqrt{\frac{GM}{r}}$.

Total mechanical energy $E = \frac{1}{2}mv^2 + \left(-\frac{GMm}{r}\right)$ 1M

$$= \frac{1}{2}m\left(\sqrt{\frac{GM}{r}}\right)^2 + \left(-\frac{GMm}{r}\right)$$

$$= -\frac{GMm}{2r}$$

$$= \frac{1}{2}U \quad \text{1M}$$

Common mistake
Students may fail to show that the total mechanical energy is $K + U$.

Revision 2 Q31 (p.90), Q32 (p.91)

Know more

Hohmann transfer orbit

Satellites may not move in a fixed orbit. They may switch from one orbit to another using *Hohmann transfer orbit*. How can this be done?

▸ p.94

Know more

Space science and modern technology NSE

Nowadays, the technology applied in space science is under rapid development. This has led to a lot of innovative inventions such as space rockets, spacecrafts and artificial satellites. The technology facilitates the study of universe as well as many other areas including resource scanning, weather monitoring, global communications, etc.

For example, our country's *Beidou Navigation Satellite System* (BDS) consists of a network of more than 30 satellites orbiting the Earth. It provides positioning, navigation and timing services on a global basis. Most of the latest mobile devices can be connected to BDS. Probably, you can enjoy the services provided by BDS right after you turn on the positioning function in your mobile phone.

Indeed, outer space security is closely related to our daily lives. How the world would be adversely affected if all the above-mentioned technologies disappeared one day?

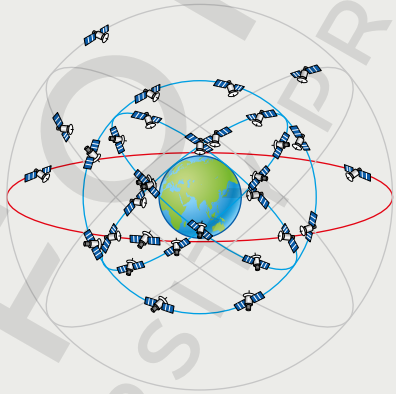


Illustration of a satellite system.

(Source: BeiDou Navigation Satellite System website in Apr 2024)

An introduction to the Beidou Navigation Satellite System is available in the website below (Chinese version).

Exercise 2.3

Take $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}.

1 A spacecraft is travelling around a planet in an elliptical orbit. The figure below shows two points X and Y on the orbit (Fig a).

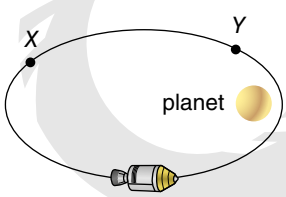


Fig a

Let U_X and U_Y be the potential energy of the spacecraft at X and Y respectively. Which of the following is correct?

A $0 > U_X > U_Y$ B $0 > U_Y > U_X$

C $U_X > U_Y > 0$ D $0 > U_X = U_Y$

2 The mass and radius of a planet are 3.4×10^{24} kg and 5.4×10^6 m respectively. What is the minimum energy needed to project an object of mass 1.5 kg from the surface of the planet to infinity?

A 12 J

B 23 J

C 6.3×10^7 J

D 1.3×10^8 J

★**3** A satellite revolves around a planet in a circular orbit. What is the relationship between the kinetic energy K and the potential energy U of the satellite?

A $K = -2U$

B $K = 2U$

C $U = -2K$

D $U = 2K$

Hohmann transfer orbit 霍曼轉移軌道 BeiDou Navigation Satellite System 北斗衛星導航系統

★**4** An object is projected from the Earth surface with a speed of $\sqrt{\frac{3GM}{R}}$ where M and R are the mass and radius of the Earth. Which of the following statements about the object is correct?

A It cannot escape from the Earth.

B Its speed is zero when it is very far away from the Earth.

C Its speed is $\sqrt{\frac{GM}{R}}$ when it is very far away from the Earth.

D Its speed is $\sqrt{\frac{2GM}{R}}$ when it is very far away from the Earth.

★**5** The radii and masses of two planets are shown in Table a.

Planet	Radius / m	Mass / kg
Mars	3.40×10^6	6.42×10^{23}
Jupiter	7.15×10^7	1.90×10^{27}

Table a

(a) Find the potential energy of a 1-kg mass at the surface of each planet.

(b) Determine which planet has a higher escape velocity according to the results in (a). Explain your answer.

★**6** A satellite moving around the Earth is transferred from its original circular orbit to another circular orbit. The radii of the original orbit and the new orbit are 7000 km and 7200 km respectively. The mass of the satellite is 550 kg and the mass of the Earth is 5.97×10^{24} kg.

(a) Find the change in potential energy of the satellite.

(b) Ginny suggests that the energy required to transfer the satellite is equal to the answer in (a). Is she correct? Explain.

★**7** (a) A satellite moves around a planet in a circular orbit of radius r. The masses of the satellite and the planet are m and M respectively. Show that the total mechanical energy of the satellite is $-\frac{GMm}{2r}$.

(b) How does the total mechanical energy of a satellite change if it moves in a higher orbit?

★**8** (a) The mass and radius of a planet is M and R respectively. Show that the escape velocity from the surface of the planet is $\sqrt{\frac{2GM}{R}}$.

(b) (i) Planet X has a mass of 1.30×10^{22} kg and a radius of 1190 km. Find the escape velocity from the surface of this planet.

(ii) Planet Y has the same radius as X but the escape velocity from the surface of Y is about double of that of X. What is the mass of Y?

★**9** A satellite moves around a planet in a circular orbit. Determine how each of the following physical quantities of the satellite changes if the satellite moves in a higher orbit. Write the equation that helps you determine.

(a) Speed

(b) Period

(c) Potential energy

(d) Kinetic energy

★**10** A satellite of mass 180 kg moves around a planet in a circular orbit with a radius of 7300 km at a speed of 8700 m s^{-1}. Then the satellite is transferred into a higher orbit of radius 7600 km (Fig b).

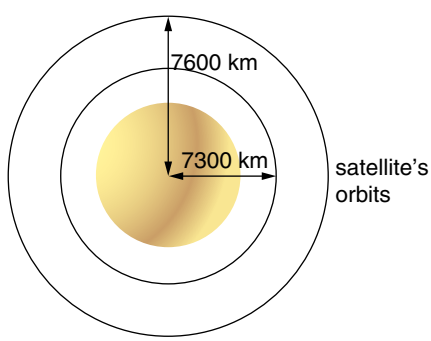


Fig b

(a) Find the mass of the planet.

(b) Find the speed of the satellite in the new orbit.

(c) Find the change in

(i) kinetic energy,

(ii) potential energy,

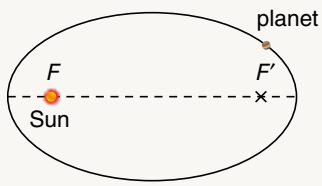
(iii) total mechanical energy

of the satellite in this orbit transfer.

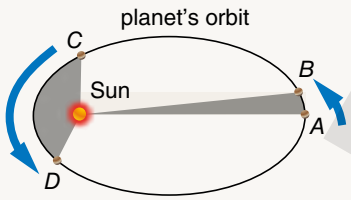
Key concepts 2

Kepler's laws of planetary motion

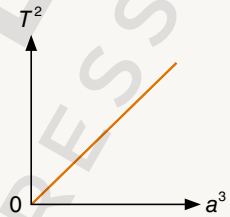
1 The planets orbit the Sun in ellipses, with the Sun at one focus.



2 The line joining the Sun and a planet sweeps out equal areas in equal time intervals.



3 $T^2 \propto a^3$
(T = orbital period;
a = semi-major axis)



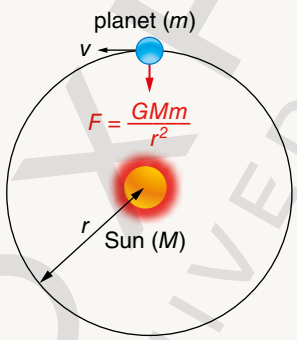
Newton's law of gravitation and Kepler's third law

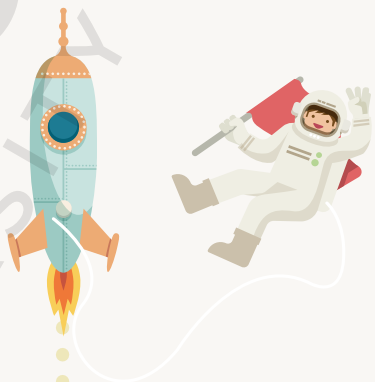
For circular orbit:

Gravitational force acting on a planet by the Sun provides the centripetal force for the circular motion of the planet.

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$T^2 = \frac{4\pi^2}{GM} r^3 \propto r^3$$





For elliptical orbit:

$$T^2 = \frac{4\pi^2}{GM} a^3 \propto a^3$$

Escape velocity

It is the minimum speed required for an object on the surface of a celestial body (of mass M and radius R) to escape from the gravitational influence of the body.

By conservation of energy,
total mechanical energy = $K + U$ = constant

$$\frac{1}{2}mv^2 + \left(-\frac{GMm}{R}\right) = 0 + 0$$

$$v = \sqrt{\frac{2GM}{R}}$$

Weightlessness

Astronauts in an orbiting spacecraft experience apparent weightlessness as both the astronants and the spacecraft fall with the same acceleration due to gravity.

Gravitational potential energy

$$U = -\frac{GMm}{r}$$

1 When $r \uparrow$,
$U \uparrow$ (less negative)

2 $U = 0$ at $r = \infty$

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Key terms

1 apparent weight 表觀重量 p.63
2 elliptical orbit 橢圓軌道 p.46
3 escape velocity 逃逸速度 p.75
4 focus (pl. foci) 焦點 p.46
5 Kepler's laws of planetary motion 開普勒行星運動定律 p.46
6 weightlessness 失重 / 無重狀態 p.63

Revision 2

Take $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}.

Multiple-choice questions

★1 2.2 Mandy and Jason experience weightlessness in a spacecraft orbiting the Earth. The mass of Mandy is smaller than that of Jason. Which of the following statements about Mandy is/are correct?

(1) The net force acting on Mandy is smaller than that on Jason.
(2) The normal force acting on Mandy is smaller than that on Jason.
(3) The acceleration of Mandy is smaller than that of Jason.

A (1) only
B (1) and (2) only
C (2) and (3) only
D (1), (2) and (3)

★2 Misc In a space mission, a satellite orbiting a planet is brought from one circular orbit to another circular orbit with a smaller radius. Which of the following quantities decreases?

A Orbital speed of the satellite
B Magnitude of gravitational force experienced by the satellite
C Potential energy of the satellite
D Minimum speed needed by the satellite to leave the planet

★3 2.1 The ratio of the semi-major axis of the orbit of Mars to that of Jupiter is 0.3 : 1. What is the ratio of their orbital periods?

A 0.11 : 1
B 0.16 : 1
C 0.32 : 1
D 0.44 : 1

★4 2.2 Which of the following statements about an astronaut in a state of apparent weightlessness is/are correct?

(1) The normal force acting on the astronaut is zero.
(2) The gravitational force is balanced by the centripetal force.
(3) The astronaut must be moving in a circular orbit.

A (1) only
B (2) only
C (1) and (2) only
D (2) and (3) only

★5 2.1 Which of the following statements about geostationary satellites is **incorrect**?

A They have the same orbital radius.
B They are equal in mass.
C Their orbital speed is constant.
D They have negative total mechanical energy.

★6 2.1 *X* is a small astronomical object orbiting the Sun. *P*, *Q*, *R* and *S* are four points on its orbit (Fig a). *X* moves slower at *P* than at *R*. At which of the following points does *X* move the fastest?

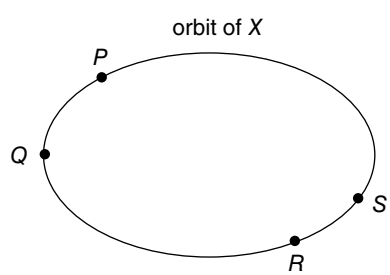


Fig a

A *Q*
B *R*
C *S*
D Cannot be determined

★7 2.3 Which of the following statements is/are correct?

(1) We can use the formula $U = -\frac{GMm}{r}$ to calculate the potential energy of an object near the Earth's surface.

(2) An object has a lower potential energy when it is further away from the Earth than when it is closer.

(3) The escape velocity from the Earth's surface will change if the potential energy of an object is taken as zero on the Earth's surface instead of at infinity.

A (1) only
B (2) only
C (1) and (3) only
D (1), (2) and (3)

★8 2.3 The escape velocity from the surface of a planet of radius R is v. What is the escape velocity from a position at $3R$ above the planet's surface?

A $\frac{1}{4}v$ B $\frac{1}{3}v$
C $\frac{1}{2}v$ D $\frac{1}{\sqrt{3}}v$

★9 2.3 A satellite can move around a planet in any circular or elliptical orbit with the planet at the centre (circular orbit) or one of the foci (elliptical orbit). Figure b shows some possible orbits.

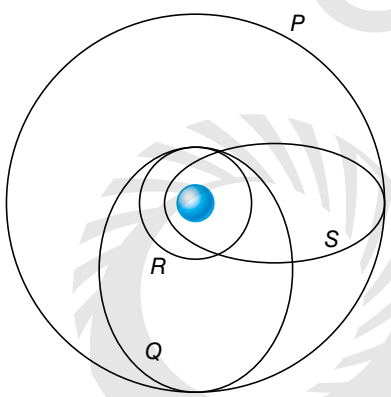


Fig b

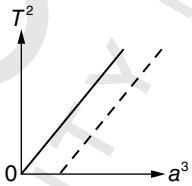
In which orbit does the satellite possess the highest total mechanical energy?

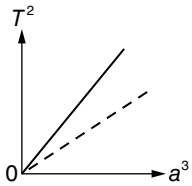
A P
B Q
C R
D S

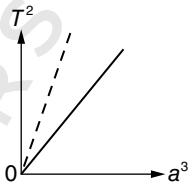
★10 2.3 A sphere of mass m is launched at a height of R_E above the Earth's surface, where R_E is the radius of the Earth. Take g as the acceleration due to gravity near the Earth's surface. If the sphere can just escape from the Earth, what is its launching speed?

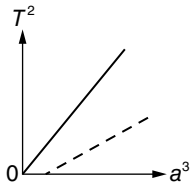
A $\sqrt{mgR_E}$ B mgR_E
C $\sqrt{2gR_E}$ D $\sqrt{gR_E}$

★11 2.1 A student plots a graph (in solid line) to investigate the relationship between the orbital periods T of the moons of Saturn and their semi-major axes a. Jupiter has a mass larger than Saturn. Which of the following graphs (in dotted line) belongs to the moons of Jupiter?

A 

B 

C 

D 

★★12 2.3 An object, initially stationary at a distance of $0.3R_E$ above the Earth's surface, disintegrates into two parts (Fig c): part X travels in a circular orbit around the Earth and part Y can just escape from the Earth. If the mass of X is m, what is the mass of Y?

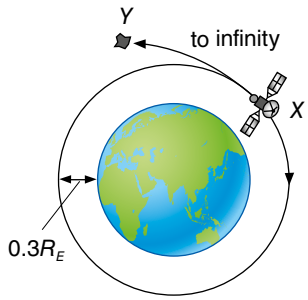


Fig c

A $\sqrt{\frac{3}{10}}m$ B $\frac{3}{10}m$
C $\frac{1}{\sqrt{2}}m$ D $\frac{1}{2}m$

▶ Refer p.71, 75

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13 ★★ 2.3 Suppose the gravitational field strength at the Earth's surface is g. A satellite of mass m is orbiting the Earth in a circular path at a height of $5R$ above the Earth's surface, where R is the radius of the Earth. What is the total mechanical energy of the satellite?

A $-\frac{1}{6}mgR$ B $-\frac{1}{10}mgR$

C $-\frac{1}{12}mgR$ D $-5mgR$

Refer p.71

14 2.3 **HKDSE 2015 Paper 2 Q1.1**

A satellite is orbiting the Earth at a distance h from the Earth's surface. What is the gain in gravitation potential energy of the satellite in the orbit with respect to the Earth's surface?

m = mass of the satellite

R = radius of the Earth

g = acceleration due to gravity on the Earth's surface

A $mgh\left(\frac{R}{R+h}\right)$ (34%)

B $mgh\left(\frac{R}{R+h}\right)^2$

C $mgh\left(\frac{R+h}{R}\right)$

D $mgh\left(\frac{R+h}{R}\right)^2$

15 Misc **HKDSE 2018 Paper 2 Q1.1**

An unpowered spacecraft is travelling in an elliptical orbit around the Earth. Which of the following descriptions is/are correct?

(1) The acceleration of the spacecraft is always pointing towards the centre of the elliptical orbit.

(2) The magnitude of the acceleration of the spacecraft is greatest when it is moving fastest.

(3) The gravitational potential energy of the spacecraft remains unchanged in the orbit.

A (1) only B (2) only (38%)

C (1) and (3) only D (2) and (3) only

16 2.3 **HKDSE 2018 Paper 2 Q1.2**

The escape velocity at the Earth's surface is 11.2 km s^{-1}. A space probe launched from the Earth's surface attains a speed of 6 km s^{-1} when it is very far away from the Earth. Find the launching speed of the space probe. Assume that the flight is unpowered except during launching and neglect the effects of other celestial bodies.

A 12.7 km s^{-1} (37%)

B 15.6 km s^{-1}

C 16.4 km s^{-1}

D 17.2 km s^{-1}

17 2.1 **HKDSE 2020 Paper 2 Q1.2**

Two satellites X and Y are revolving in clockwise direction about the Earth as shown. The diameter of the circular orbit of X equals the length of the major axis of the elliptical orbit of Y. The two orbits intersect at P and Q.

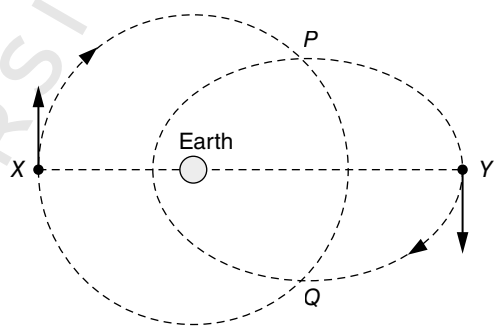


Fig d

At the instant shown, the two satellites and the Earth are on a straight line. Which deductions are correct?

(1) X and Y have the same acceleration when they pass P.

(2) The speed of X is greater than that of Y at the instant shown.

(3) The satellites will not meet each other at P or at Q.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3) (24%)

Conventional questions

★**18** 2.1 The orbital periods T and the semi-major axes a of some satellites around planet P are measured. The graph below shows the variation of T^2 against a^3 (Fig e).

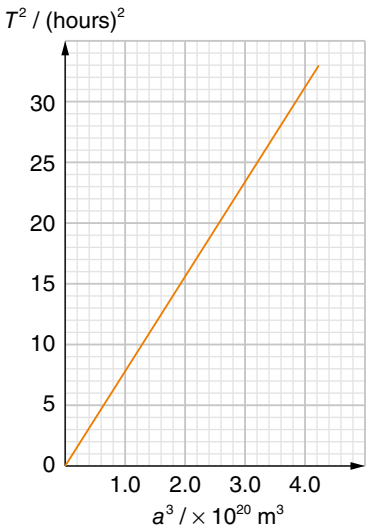


Fig e

(a) Explain how the motion of the satellites satisfies Kepler's third law. (2 marks)

(b) Estimate the mass of planet P. (3 marks)

(c) A satellite with an orbital period of 8.5 hours and a semi-major axis of 6×10^6 m is found. Does this satellite orbit planet P? Explain briefly. (3 marks)

★**19** Misc A communications satellite moves in a geostationary orbit around the Earth (Fig f). Take the mass of the Earth to be 5.97×10^{24} kg.



Fig f

(a) What is a geostationary orbit? (1 mark)

(b) Find the radius of the orbit. (2 marks)

(c) Find the orbital speed of the satellite. (2 marks)

(d) The mass of the satellite is 1000 kg. Find its mechanical energy. (2 marks)

★**20** 2.1 Figure g shows the orbit of a comet around the Sun S. O is the centre of the orbit. A, B, C and D are four points on the orbit. It is known that OS = 2 AU and SA = 1.5 AU.

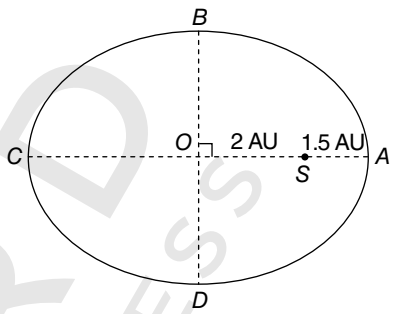


Fig g

(a) Find the furthest distance of the comet from the Sun. (1 mark)

(b) Find the orbital period (in years) of the comet. Take the Earth's semi-major axis to be 1 AU. (3 marks)

(c) The arc length of AB is equal to that of CD. Along which arc does the comet travel in a shorter time, AB or CD? Explain briefly. (2 marks)

★**21** Misc A spacecraft is near the surface of Jupiter and is moving with an acceleration of 25.9 m s^{-2} downwards. The mass and radius of Jupiter are 1.90×10^{27} kg and 70 000 km respectively.

(a) Find the acceleration due to gravity near the surface of Jupiter. (2 marks)

(b) Is the passenger inside the spacecraft experiencing weightlessness? Explain your answer. (2 marks)

(c) The spacecraft ejects a device upwards. What is the minimum speed (relative to Jupiter) of the device so that it can escape from Jupiter? (2 marks)

★**22** 2.1 A communications satellite can be placed in an orbit so that it remains directly above the same location on the Earth all the time. Determine and explain whether each of the following statements is correct.

(a) The satellite must be far away from the Earth so that it is not affected by the Earth's gravity. (2 marks)

(b) Hong Kong is a possible location for this kind of satellite. (2 marks)

(c) This kind of satellite must all have the same constant speed. (3 marks)

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★**23** 2.3 Europa is a satellite of Jupiter. The radius of Europa is 1569 km and the acceleration due to gravity at its surface is 1.31 m s^{-2}.

(a) Calculate the escape velocity from Europa's surface. (3 marks)

(b) Calculate the mass of Europa. (2 marks)

(c) A body is projected vertically upwards from the surface of Europa with a speed equal to half of the escape velocity. Find the maximum height it can reach. (3 marks)

(d) In practice, the speed required for an object to escape from the solar system is larger than that in (a). Explain briefly. (2 marks)

★**24** Misc Comet Halley orbits the Sun with a period of 75.3 years (Fig h). Its shortest distance d from the Sun is 8.78×10^{10} m. Take the orbital period of the Earth and the semi-major axis of its orbit to be 1 year and 1.50×10^{11} m, and the mass of the Sun to be 1.99×10^{30} kg.

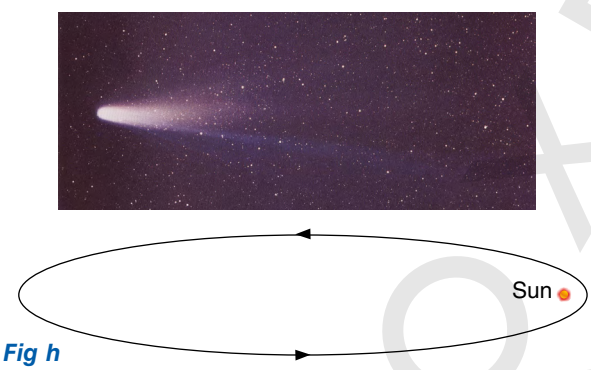


Fig h

(a) Find the semi-major axis of the comet's orbit. (2 marks)

(b) Hence find its longest distance from the Sun. (1 mark)

(c) The comet has a speed of 54.6 km s^{-1} at its closest approach.

(i) The total length of Comet Halley's orbit is about 1.22×10^{10} km. Estimate the average speed (in km s^{-1}) of Comet Halley in completing one orbit. Take 1 year = 3.16×10^7 s. (1 mark)

(ii) By considering Kelper's law, explain why the result in (c)(i) is much lower than 54.6 km s^{-1}. (2 marks)

(iii) Find the speed of Comet Halley when it is at a distance of $2d$ from the Sun. (2 marks)

★**25** 2.3 A spacecraft of mass 8130 kg was launched to move in a circular orbit around the Earth. Take $GM = 4.0 \times 10^{14}$ N m^2 kg^{-1}, where M is the mass of the Earth.

(a) What keeps the spacecraft orbiting around the Earth? (1 mark)

(b) The potential energy of the spacecraft is -4.83×10^{11} J. Find the orbital radius of the spacecraft. (2 marks)

(c) Find the total mechanical energy of the spacecraft. (3 marks)

(d) An object is projected from the spacecraft. Assume the object is not powered and does not collide with the Earth after launch. Find the launching speed of the object if

(i) the object can just escape from the Earth's gravitational attraction;

(ii) the object has a speed of 2 km s^{-1} when it is very far away from the Earth.

(4 marks)

★**26** 2.3 An object X of mass m moves around a planet of mass M in a circular orbit of radius r. The radius of the planet is 1.0×10^6 m. If X is at the surface of the planet, its potential energy will be -4×10^6 J.

(a) Show that the kinetic energy of X in its orbit is $\frac{GMm}{2r}$. (2 marks)

(b) Given that the kinetic energy of X in its orbit is 1.2×10^6 J, what is its potential energy in its orbit? (2 marks)

(c) The mass of X is 2 kg. Find the mass of the planet. (2 marks)

(d) If X needs to escape from the planet, extra energy should be provided to it. Find the minimum extra energy needed. (2 marks)

★**27** Misc The total mechanical energy of a small object of mass m in an elliptical orbit around a celestial body of mass M can be generalized as:

$$E = -\frac{GMm}{2a}$$

where a is the semi-major axis of the elliptical orbit.

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(a) (i) Show that when the small object is at a distance d from the celestial body, the speed of the small object is given by

$$v = \sqrt{GM\left(\frac{2}{d} - \frac{1}{a}\right)}.$$ (2 marks)

(ii) From the formula in (a)(i), derive the formula for the orbital speed of a small object moving in a circular orbit of radius r around a celestial body of mass M. (2 marks)

(b) Figure i shows an elliptical orbit with semi-major axis a and a circular orbit of radius r around the same celestial body, with $a = 2r$. Object X moves in the elliptical orbit while object Y moves in the circular orbit. When X is at a certain position in the elliptical orbit, its speed is the same as Y. Write the condition satisfied by that position. Show your calculation. (2 marks)

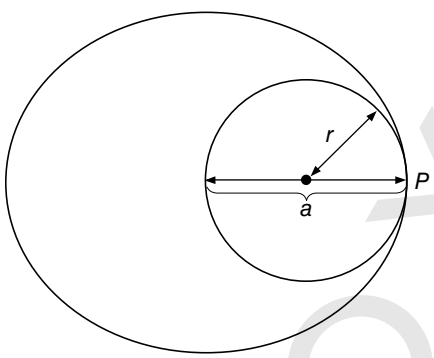


Fig i

★ **28** Misc

A spacecraft, 2001 Mars Odyssey, was sent from the Earth to orbit Mars in 2001 (Fig j). Initially, the spacecraft moved in an elliptical orbit with a period of 19.9 hours (Fig k). It is closest to Mars at point X, which is 400 km above the surface of Mars. Then it gradually loses energy and its orbit changes from elliptical to circular.



Fig j

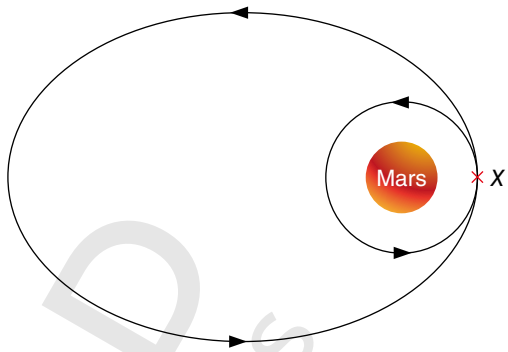


Fig k

The mass and radius of Mars are 6.42×10^{23} kg and 3390 km respectively.

(a) Find the semi-major axis of the elliptical orbit. (2 marks)

(b) Find the period of the spacecraft in the circular orbit. (2 marks)

(c) Find the speed of the spacecraft in the circular orbit. (2 marks)

(d) (i) The total mechanical energy and the potential energy of the spacecraft in the circular orbit are E and U respectively.

Show that $E = \frac{1}{2}U$. (2 marks)

(ii) Show that the spacecraft has a lower energy in the circular orbit than in the elliptical orbit. (2 marks)

★★ **29** Misc

In the solar system, there are eight planets orbiting around the Sun (Fig l).

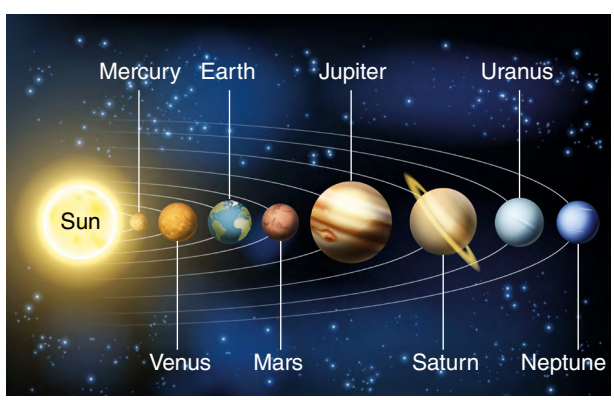


Diagram not drawn to scale

Fig l

Take the orbital period of the Earth and the semi-major axis of its orbit to be 1 year and 1 AU, the mass and radius of the Earth to be 5.97×10^{24} kg and 6370 km respectively.

(a) Which planet has the longest orbital period? Explain your answer. (2 marks)

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(b) The orbital period of Saturn is about 30 years. Estimate the semi-major axis (in AU) of its orbit. (2 marks)

(c) The International Space Station (ISS) orbits around the Earth in a nearly circular orbit at a height of 402 km above the Earth's surface. Its mass is about 450 000 kg. An astronaut working in ISS has a mass of 70 kg and experiences weightlessness.

(i) Explain why the astronaut experiences weightlessness. (1 mark)

(ii) What is the weight of the astronaut? (2 marks)

(iii) The Earth can be considered as a huge 'space station' orbiting the Sun, but we do not feel weightlessness when standing on the Earth's surface. Why? (2 marks)

Refer p.64–65

★★ **30** Misc

Figure m shows the orbits of two objects X and Y around the Sun. The orbits of X and Y are circular and elliptical respectively, and the two orbits intersect at P. O is the mid-point of the two foci of the elliptical orbit. OP is the diameter of the orbit of X.

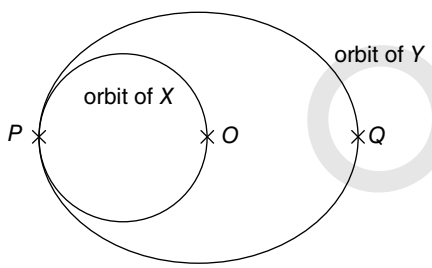


Fig m

(a) Indicate the position of the Sun in Figure m. Hence, state where Y attains its maximum speed. (2 marks)

(b) (i) Write down the relationship between the radius r of the orbit of X and the semi-major axis a of the orbit of Y. (1 mark)

(ii) Hence, derive the relationship between the periods of X and Y. (2 marks)

(c) Which object would have a complete cycle of phases as observed from the other one? Why? (2 marks)

Refer p.30, 56, 58

See p.92 for the Exam Report.

31 Misc

HKDSE 2013 Paper 2 Q.1

Given: $GM = 4.0 \times 10^{14}$ N m^2 kg^{-1}, where G is the universal gravitational constant and M is the mass of the Earth.

Mean radius of the Earth = 6400 km.

Radius of the geostationary orbit is about 42 400 km, i.e. 36 000 km above the Earth's surface.

The following describes a way to launch a satellite into the geostationary orbit:

- The satellite is first launched by a rocket to a circular near-Earth orbit (1) at 300 km above the Earth's surface.
- At A, the satellite's engine is fired for a short period of time to give it a boost needed to enter the elliptical transfer orbit (2), with AB as the ellipse's major axis.
- At B, the satellite's engine is fired again briefly to boost it into the geostationary orbit (3).

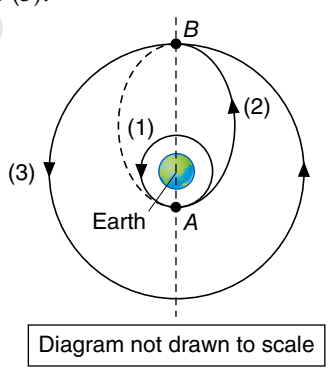


Fig n

Assume that the three orbits are coplanar such that the elliptical orbit touches the two circular orbits at A and B respectively. During the period when the satellite travels from A to B along the transfer orbit, its engine is shut.

(a) Communications satellites are usually launched into the geostationary orbit. State and explain the advantage of such an arrangement. (2 marks)

(b) Find the speed of the satellite in the near-Earth orbit (1). (2 marks)

(c) (i) Show that for a satellite of mass m moving in a circular orbit of radius r around the Earth, its total mechanical energy is $-\frac{GMm}{2r}$, where M is the mass of the Earth. Take the gravitational potential energy of the satellite at infinity to be zero. (2 marks)

(ii) Use the result in (c)(i) to calculate the energy required to transfer a satellite of mass m = 2000 kg from the near-Earth orbit (1) through A to the geostationary orbit (3) through B. (2 marks)

(iii) How long does it take for the satellite to travel from A to B along the transfer orbit (2)? (2 marks)

32 **HKDSE 2019 Paper 2 Q.1**

Misc

Figure o shows a space station S revolving in a circular orbit at a height of 400 km **above** the Earth's surface.

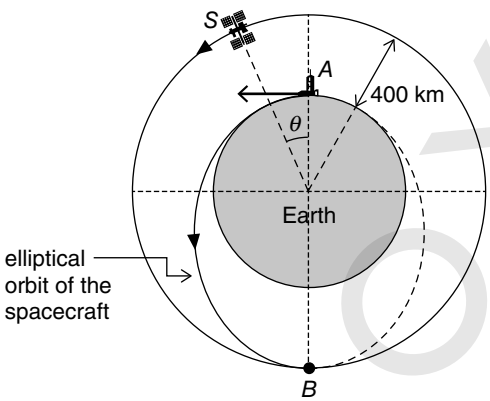


Diagram not drawn to scale

Fig o

A spacecraft is launched with speed 8.02 km s^{-1} from A at the Earth's surface to meet the station S through an elliptical orbit with AB as the major axis. The spacecraft's rocket engine is shut when it coasts from A to B along the elliptical orbit. Assume that the two orbits are in the same plane.

Given: $GM = 4 \times 10^5$ km^3 s^{-2}, where G is the universal gravitational constant and M is the mass of the Earth.

Radius of the Earth = 6400 km

(a) (i) Using conservation of total mechanical energy, or otherwise, find the speed v_B of the spacecraft when it reaches B. Neglect the effects of the atmosphere. (2 marks)

(ii) Show that the spacecraft takes about 2663 s to travel from A to B. (2 marks)

(iii) Explain why an astronaut in the orbiting spacecraft would experience 'weightlessness'. (1 mark)

(b) The space station S travels at a constant speed of 7.67 km s^{-1} in the circular orbit with a period of 5570 s.

(i) If the spacecraft is to meet the station S exactly when it reaches B, use the result in (a)(ii) to show that their angular separation θ (shown in Figure o) when the spacecraft has just launched at A should be slightly less than 8°. (2 marks)

(ii) In order to make the spacecraft's speed v_B found in (a)(i) exactly the same as that of the station S when they meet at B, a student suggests to slightly adjust the launching speed of the spacecraft at A. Comment on the feasibility of the suggestion. (2 marks)

(iii) Suggest one simple way for the spacecraft at B to travel with the same speed as station S. (1 mark)

Q32 Exam Report: In (a)(i), most candidates knew how to find the speed of the spacecraft at B using conservation of total mechanical energy though a few made mistakes in unit conversion. Some candidates failed to answer (a)(ii) correctly as they did not know that the semi-major axis of the elliptical orbit should be employed in the calculation. Not many were able to explain the 'weightlessness' phenomenon in (a)(iii). A few had a misconception that both the astronaut and the spacecraft moving at the same acceleration would necessarily result in weightlessness. Quite a number of candidates managed to find the angular separation θ required in (b)(i) using various methods. Part (b)(ii)(iii) were poorly answered. It seemed that most candidates did not know that the initial launching speed of the spacecraft at the Earth's surface would eventually determine the shape of its trajectory.

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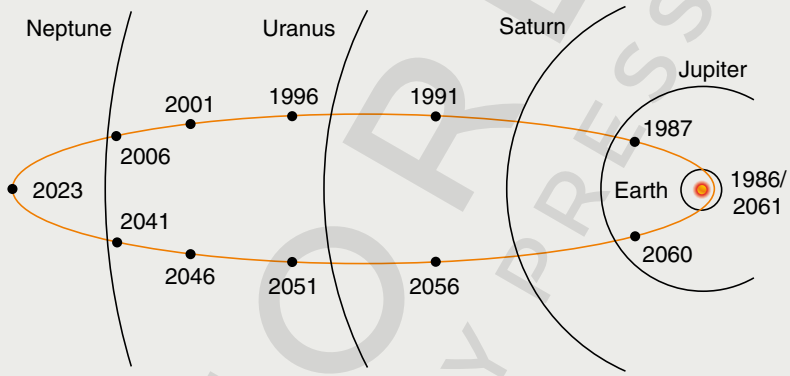
Q31 Exam Report: In (a), some candidates were not familiar with the geostationary orbit and the applications of the satellites in it. Part (b) was well answered although mistakes like substituting incorrect radii, missing square roots or using wrong units were common. In (c)(i), the less able candidates did not demonstrate they understood that total mechanical energy is the sum of kinetic energy and potential energy. They failed to use the result in (c)(i) to attempt (c)(ii) or wrongly stated that $\Delta E = \frac{-GMm}{2}\left(\frac{1}{r_1 - r_2}\right)$. A few candidates omitted the factor $\frac{1}{2}$ or gave a negative ΔE. In (c)(iii), quite a number of candidates failed to find the semi-major axis while they forgot that only half of the period was required for the transfer orbit (2).

Know more

Comet Halley (p.53)

Comet Halley is one of the comets that can be seen clearly with the naked eye from the Earth. It is a short-period comet with an orbital period of about 75 years. Other comets may have orbital periods about thousands of years.

Compared with the planets in the solar system, the orbit of Comet Halley is highly elliptical. Its perihelion at 0.586 AU is located between the orbits of Mercury and Venus, while its aphelion at 35 AU is located beyond Neptune's orbit.



The orbit of Comet Halley. Each number represents the comet's position in that year.

Kepler's second law is especially noticeable for celestial bodies with highly elliptical orbits such as comets. Comet Halley sweeps quickly (within months) through the inner part of the solar system, and spends the rest of its 75-year period moving slowly far away from the Sun.

Writing in space (p.65)

The space pen, also known as the zero gravity pen, is a pen which uses pressurized ink cartridges to push the ink to the tip. The pen can write from all directions, in zero gravity and in extreme temperatures.



An astronaut writing with a space pen in space.



A pencil is easily broken and produces graphite dust.

Before the space pen was invented, astronauts used pencils to write. However, the soft lead core of a pencil broke easily during writing. The graphite dust produced in writing can conduct electricity. The dust floating around in the spacecraft may cause short circuits in the equipment, which can be fatal.

Simulation

→ This simulation shows how the path of a body is affected by the launching speed. It is available on OUP web in PC version (.exe) only.

Know more

The apple, the Moon and Newton's cannonball (p.59)

In order to illustrate that the Moon could move under the same force that pulled an apple towards the ground, Newton made up a thought experiment: imagine a cannonball being fired horizontally from the top of a very high mountain with speed v (Fig a). If air resistance is negligible, the cannonball falls freely under gravity after leaving the cannon.

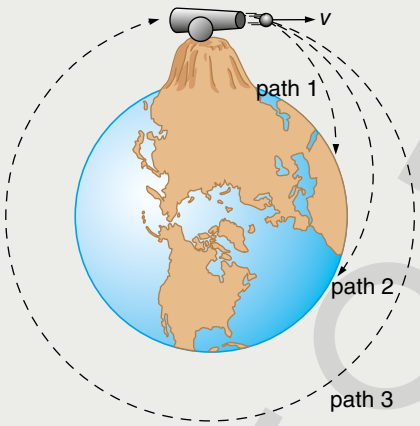


Fig a

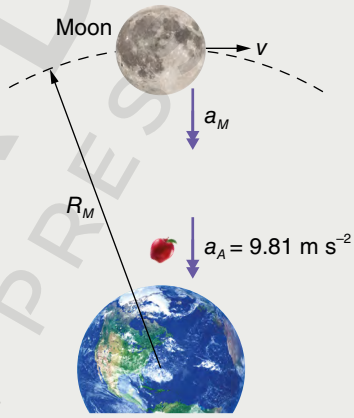


Fig b

The cannonball would usually hit the ground after travelling a certain distance. The higher the speed v, the further away the cannonball lands (paths 1 and 2).

When v reaches a certain value, the cannonball would fall at the same rate as the ground curves. It could then keep moving at the same height above the ground and would never hit the ground. In other words, the cannonball moves in a circular orbit around the Earth (path 3).

The Moon orbiting the Earth is in a similar case as the cannonball flying in path 3. We can then regard the Moon's motion in the same way as the free-falling of an apple (Fig b).

Newton knew that the Moon took about 27.3 days to orbit once around the Earth, and the radius R_M of the Moon's orbit is about 60 times the radius of the Earth (known to be 6.37×10^6 m). He also knew that the acceleration of a free-falling apple is 9.81 m s^{-2}. The Moon's acceleration a_M can be calculated as follows.

$$a_M = \frac{v^2}{R_M} = \frac{\left(\frac{2\pi R_M}{T}\right)^2}{R_M} = \frac{4\pi^2 R_M}{T^2} = \frac{4\pi^2(60 \times 6.37 \times 10^6)}{(27.3 \times 24 \times 60 \times 60)^2} = 2.71 \times 10^{-3}\ \text{m s}^{-2}$$

$$\frac{a_M}{a_A} = \frac{2.71 \times 10^{-3}}{9.81} \approx \frac{1}{3600} = \left(\frac{1}{60}\right)^2 = \left(\frac{R_E}{R_M}\right)^2$$

$$\therefore a \propto \frac{1}{r^2}$$

This supports Newton's thought that the acceleration decreases as the square of distance, thus providing evidence that the same force (obeying an inverse-square law) keeps the Moon in orbit and pulls an apple towards the ground.

Know more

The atmosphere (p.75)

Lighter gas molecules, like hydrogen and helium, move at a higher average speed than heavier ones, like oxygen and nitrogen. Some energetic molecules may reach a speed higher than the escape velocity of a celestial body and thus escape from it. That is one reason why there is less hydrogen and helium in the Earth's atmosphere.

For the Moon, the escape velocity is smaller, resulting in a much thinner atmosphere than the Earth.



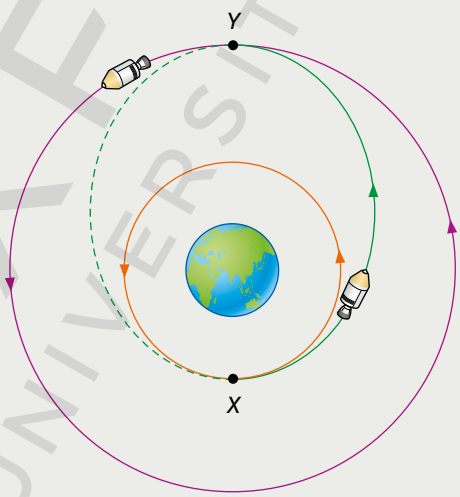
The Earth and its atmosphere.

In general, we compare the root-mean-square speed of a gas, which is given by $\sqrt{\frac{3RT}{N_A m}}$ (see Book 1 Chapter 5), with the escape velocity of a celestial body to predict the presence of the gas on that celestial body.

✎ The Moon has a very thin atmosphere with gas composition different from that of the Earth. Check it out on the ESA website below.

Hohmann transfer orbit (p.81)

Satellites may not move in a fixed orbit. They may switch from one orbit to another using Hohmann transfer orbit.



Consider a simple orbit transfer from a smaller circular orbit (orange path) to a larger circular orbit (violet path) around the Earth. The transfer can be done as follows:

1. The spacecraft switches on its engine at *X* to increase its speed, so that it enters an elliptical orbit around the Earth (i.e. from orange path to green path).
2. When the spacecraft reaches *Y*, which is the point of furthest approach of the elliptical orbit, it increases its speed again so that the spacecraft enters the desired orbit (i.e. from green path to violet path).

The part of the elliptical orbit *XY* taken during the orbit transfer is called Hohmann transfer orbit.

The orbit transfer can be reversed by decreasing the speed of the spacecraft at *Y* and *X*, so that the spacecraft enters the Hohmann transfer orbit and then back to the original smaller circular orbit.

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